

Research Methods in Computer Science

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Research Methods in Computer Science

Lecture 1: Introduction and Overview

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Today ...

1 Introduction and Overview

- Aims
- Learning outcomes
- Delivery
- Assessment

2 What is 'Research'?

Aims

- ➊ To provide a deep and systematic understanding of the nature and conduct of Computer Science research
- ➋ To equip students with the ability to undertake independent research
- ➌ To enhance existing transferable key skills
- ➍ To develop high-order transferable key skills
- ➎ To remind students of the Legal, Social, Ethical and Professional (LSEP) issues applicable to the computer industry

Learning Outcomes (1)

- ① Have an **understanding** of how established **techniques of research** and enquiry are used to extend, create and interpret knowledge in Computer Science
- ② Have a conceptual **understanding sufficient to**:
 - (i) **evaluate** critically **current research** and advanced scholarship in Computer Science, and
 - (ii) **propose** possible **alternative directions** for further work
- ③ Be **able to deal with complex issues** at the forefront of the academic discipline of Computer Science in a manner, based on sound judgements, that is both systematic and creative; and be **able to communicate conclusions clearly** to both specialists and non-specialists

Learning Outcomes (2)

- 4 Demonstrate self-direction and originality in tackling and solving problems within the domain of Computer Science, and be able to act autonomously in planning and implementing solutions in a professional manner
- 5 Be able to define and plan a programme of independent research
- 6 Participate within the professional, legal and ethical framework within which they would be expected to operate as professionals within the IT industry

Learning Outcomes (3)

- 7 Make use of the qualities and **transferable skills** necessary for employment requiring:
 - (i) the exercise of **initiative** and **personal responsibility**,
 - (ii) **decision making in complex and unpredictable situations**, and
 - (iii) the **independent learning ability** required for continuing professional development
- 8 Have the **skills set** to be able to continue to **advance their knowledge and understanding**, and to **develop new skills to a high level**, with respect to continuing professional development as a “**self-directed life-long learner**” across the discipline of Computer Science

Learning Outcomes (4)

In short, you should learn

- 1 to **understand research** and **research methods** in Computer Science;
- 2 to **be able to plan**, and **conduct your own research**, taking into account ethical, legal, and professional limitations; and
- 3 to **be able to communicate its results**

Delivery of the module (1)

1 Lectures

Monday,	11.00	Ashton Building Seminar Room (Ground Floor)
Monday,	12.00	Ashton Building Seminar Room (Ground Floor)
Friday,	10.00	Ashton Building Seminar Room (Ground Floor)

2 Seminars

During lectures	By members of staff
Tuesday, 16.00	Departmental research seminar Ashton Building Seminar Room (Ground Floor)

3 Tutorials/Labs

To be arranged

Delivery of the module (2)

① Office hours

Typically, Monday 15.00 to 17.00;
make an arrangement by e-mail (U.Hustadt@liverpool.ac.uk) first

② Website

<http://www.csc.liv.ac.uk/~ullrich/COMP516/>

Recommended texts

- Christian W. Dawson: Projects in Computing and Information Systems (A Student's Guide). Addison Wesley, 2005.
Harold Cohen Library, Class No 518.561.D27

Earlier edition:
Christian W. Dawson: The essence of computing projects (A student's guide). Prentice Hall, 2000.
Harold Cohen Library, Class No 518.561.D27
- Justin Zobel: Writing for Computer Science. Springer, 2004.
Harold Cohen Library, Class No 378.962.Z81

Assessment

- 2,000 word essay on a topic chosen by the examiner
 - to be handed out on Friday, 28 September 2007
 - to be submitted on Friday, 26 October 2007, 15.30
 - accounts for 15% of the module mark
- 5,000 word essay on an agreed subject
 - work can be started as soon as the subject is agreed (Friday, 26 October 2007, at the latest)
 - to be submitted before the Xmas break (most likely Thursday, 13 December 2007, 15.30)
 - accounts for 85% of the module mark

Pass mark, as usual for MSc modules, is 50%

Teaching and learning strategy

- Lectures, seminars, tutorials/labs only make up a small part of the delivery of the module
- In total you are expected to commit 150 hours to the module, that is, **12.5 hours** per week over 12 weeks (more hours per week than for any other module)
- Of those the timetabled activities only make up **4 hours per week**
- In addition you should spend **3.5 hours per week** on reflection, consideration of lecture material and background reading plus **5 hours per week** on the **assessment tasks**

What is 'research'?

Research (Dictionary)

Noun

- 1 Scholarly or scientific investigation or inquiry.
- 2 Close, careful study.

Verb

- 1 To study (something) thoroughly so as to present in a detailed, accurate manner.
(Example: researching the effects of acid rain.)

Note the difference between the definition of the noun and of the verb.

What is 'research'?

Study (Dictionary)

Noun

- ① The pursuit of knowledge, as by reading, observation, or research.
- ② Attentive scrutiny.

Verb

- ① To apply one's mind purposefully to the acquisition of knowledge or understanding of (a subject).
- ② To inquire into; investigate.
- ③ To examine closely; scrutinise.

What is 'research'?

Research (<http://en.wikipedia.org/wiki/Research>)

- an active, diligent, and systematic process of inquiry in order to discover, interpret or revise facts, events, behaviours, or theories, or to make practical applications with the help of such facts, laws, or theories.
- a collection of information about a particular subject.
- derives from the Middle French and the literal meaning is “to investigate thoroughly”.

Homework: Read the Wikipedia article!

Research Methods in Computer Science

Lecture 2: Research (continued)

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Previously ...

1 Introduction and Overview

- Aims
- Learning outcomes
- Delivery
- Assessment

2 What is 'Research'?

Today ...

3 What is 'Research'?

- More Definitions of 'Research'

4 Knowledge

- A Hierarchy
- Data
- Information
- Knowledge

5 Knowledge

- Theories

6 Originality

- Definition
- The importance of repeating the work of others

What is 'research'?

Research (<http://en.wikipedia.org/wiki/Research>)

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Homework: Read the Wikipedia article!

What is 'research'?

Research (Higher Education Funding Council for England)

Original investigation undertaken in order to gain knowledge and understanding, including

- work of direct relevance to the needs of commerce and industry and to the public and voluntary sectors
- scholarship (research infrastructure)
- the invention and generation of ideas, images, performances and artifacts including design, where these lead to new or substantially improved insights;
- the use of existing knowledge in experimental development to produce new or substantially improved materials, devices, products and processes, including design and construction.

Knowledge: A hierarchy

Knowledge is a particular level in a hierarchy:

- 1 Data
- 2 Information
- 3 Knowledge
- 4 [Wisdom]

Knowledge: Data and Information

Datum/Data

- statements accepted at face value (a 'given') and presented as numbers, characters, images, or sounds.
- a large class of practically important statements are **measurements** or **observations** of variables, objects, or events.
- in a computing context, in a form which can be **assessed**, **stored**, **processed**, and **transmitted** by a computer.

Knowledge: Data and Information

Information

- **Data** on its own has no meaning, only when **interpreted** by some kind of **data processing system** does it take on meaning and becomes **information**

Example:

The **human genome project** has determined the sequence of the 3 billion chemical base pairs that make up human DNA

→ identifying base pairs produces **data**

→ **information** would tell us what they do!

Knowledge: Alternative definitions (1)

Knowledge (Dawson 2005)

- higher level understanding of things
- represents our understanding of the 'why' instead of the mere 'what'
- interpretation of information in the form of rules, patterns, decisions, models, ideas, etc.

In **natural sciences**, understanding 'why' is too ambitious most of time; understanding 'how' is usually what we aim for

In other areas, understanding 'how' is trivial, understanding 'why' is challenging

Knowledge: Alternative definitions (2)

Knowledge (Davenport et al. 1998)

- a fluid mix of **framed experience**, **contextual information**, values and **expert insight** that provides a **framework for evaluating and incorporating new experiences and information**.
- information combined with experience, context, interpretation, and reflection
- high-value form of information that is ready to apply to decisions and actions

Second point similar to last point in the previous definition

Last point seems to imply that knowledge has to be useful
(is astrophysics useful?)

Knowledge: Alternative definitions (3)

Knowledge (<http://en.wikipedia.org/wiki/Knowledge>)

- the **awareness** and **understanding of facts**, truths or information gained in the form of experience or learning (a posteriori), or through deductive reasoning (a priori)
- an appreciation of the possession of **interconnected details** which, in isolation, are of lesser value
- both knowledge and information consist of true statements, but knowledge is information that has a **purpose or use** (information plus intentionality)

Knowledge and theories: Definition

Scientific knowledge is often organised into **theories**.

Theory (<http://en.wikipedia.org/wiki/Theories>)

- a **logically self-consistent model** or framework describing the behaviour of a certain natural or social phenomenon, thus either originating from observable facts or supported by them
- formulated, developed, and evaluated according to the **scientific method**

Knowledge and theories: Criteria

Theory (<http://en.wikipedia.org/wiki/Theories>)

A body of (descriptions of) knowledge is usually only called a **theory** once it has a **firm empirical basis**, that is, it

- 1 is **consistent with pre-existing** theory to the extent that the pre-existing theory was experimentally verified, though it will often show pre-existing theory to be wrong in an exact sense,
- 2 is **supported by many strands of evidence** rather than a single foundation, ensuring that it probably is a good approximation if not totally correct,

Knowledge and theories: Criteria

Theory (<http://en.wikipedia.org/wiki/Theories>)

A body of (descriptions of) knowledge is usually only called a **theory** once it has a **firm empirical basis**, that is, it

- ③ **makes (testable) predictions** that might someday be used to disprove the theory, and
- ④ has **survived many critical real world tests** that could have proven it false,
- ⑤ is a/the **best known explanation**, in the sense of Occam's Razor, of the infinite variety of alternative explanations for the same data.

Knowledge and theories: Facts versus theories

‘This (e.g. evolution) is only a **theory** not a **fact**’

Fact

1. a **truth** (statement confirming to **reality**)
or
2. **data** supported by a **scientific experiment**

- Status of a ‘**truth**’ is by and large unachievable
- A **theory** is formulated, developed, and evaluated according to the **scientific method**

Given enough **experimental support** a **theory** can be
(a scientific) **fact**

Originality (1)

Research (HEFCE): Original investigation undertaken in order to gain knowledge and understanding

Originality

Doing something that has not been done before

Dawson (2005):

There is no point in repeating the work of others and discovering or producing what is already known

Only true for what is truly known (i.e. very little)

- Theories make predictions, which need to be tested
- The people performing those tests are neither infallible nor trustworthy
- Tests need to be repeated and results replicated

(In)Fallibility

Cold fusion

(http://en.wikipedia.org/wiki/Cold_fusion)

- **Cold fusion**: Nuclear fusion reaction that occurs well below the temperature required for thermonuclear reactions, that is, near ambient temperature instead of millions of degrees Celsius
- First reported to have been achieved by Pons (University of Utah) and Fleischmann (University of Southampton) in 1989
- Scientists tried to replicate their results shortly after initial announcement
- Teams at Texas A&M University and the Georgia Institute of Technology first confirmed the results, but then withdraw those claims due to lack of evidence
- Vast majority of experiments failed

Research Methods in Computer Science

Lecture 3: Research (continued)

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Previously ...

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- More Definitions of 'Research'

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- a collection of information about a particular subject.

Research (Higher Education Funding Council for England)

Original investigation undertaken in order to gain knowledge and understanding

Today ...

- 7 Investigation
- 8 Knowledge
- 9 Originality
 - Areas of originality
- 10 Gain
- 11 What is 'Research' ?
 - Summary

Investigation

‘An active, diligent, and systematic process of inquiry’ (Wikipedia)

- Scientists use observations and reasoning to develop technologies and propose explanations for natural phenomena in the form of hypotheses
- Predictions from these hypotheses are tested by experiment and further technologies developed
- Any hypothesis which is cogent enough to make predictions can then be tested reproducibly in this way
- Once it has been established that a hypothesis is sound, it becomes a theory.
- Sometimes scientific development takes place differently with a theory first being developed gaining support on the basis of its logic and principles

Knowledge: A hierarchy

Datum/Data

- statements accepted at face value (a 'given') and presented as numbers, characters, images, or sounds.
- a large class of practically important statements are **measurements** or **observations** of variables, objects, or events.

Information

- **Data interpreted** by some kind of **data processing system** which gives it meaning

Knowledge (Dawson 2005)

- higher level understanding of things
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Research and Originality (1)

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- Tests need to be repeated and results replicated

Research and Originality (2)

Areas of originality (Cryer 1996)

- Exploring the unknown
Investigate a field that no one has investigated before
- Exploring the unanticipated
Obtaining unexpected results and investigating new directions in an already existing field
- The use of data
Interpret data in new ways
- Tools, techniques, procedures, and methods
Apply new tools/techniques to alternative problems
Try procedures/methods in new contexts

Gain

Research (HEFCE): Original investigation undertaken in order to gain knowledge and understanding

Contribution

Research is supposed to add to the world's body of knowledge and understanding (in contrast to adding to the researcher's knowledge and understanding)

What is 'research'?

In summary, what are the **three key aspects of research**?

(10 minutes group discussion)

What is 'research'?

Research (<http://en.wikipedia.org/wiki/Research>)

An active, diligent, and systematic process of inquiry in order to discover, interpret or revise facts, events, behaviours, or theories, or to make practical applications with the help of such facts, laws, or theories.

Research (Higher Education Funding Council for England)

Original investigation undertaken in order to gain knowledge and understanding

Sharp et al. (2002)

Seeking through methodical process to add to one's own body of knowledge and to that of others, by the discovery of non-trivial facts and insights

What is 'research'?

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What is 'research'? (Summary)

Wikipedia	HEFCE	Sharp
active, diligent, and systematic process of inquiry	investigation	methodical process
discover, interpret, or revise		discovery
	gain	add
facts, events, behaviours, or theories	knowledge and understanding	knowledge / non-trivial facts and insights

Research Methods in Computer Science

Lecture 4: Research process models

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Previously ...

- 7 Investigation
- 8 Knowledge
- 9 Originality
 - Areas of originality
- 10 Gain
- 11 What is 'Research' ?
 - Summary

Topics

12 Research process models

- Sequential
- Generalised
- Circulatory
- Evolutionary

Research process models

All definitions agree that **research** involves a **systematic** or **methodical** process

Dawson (2005), following Baxter (2001), identifies four common views of the **research process**:

- Sequential
- Generalised
- Circulatory
- Evolutionary

Research process models: Sequential (1)

Research process as

- Series of activities
- Performed one after another (sequentially)
- In a fixed, linear series of stages

Example:

Research process model of Greenfield (1996):

- 1 Review the field
- 2 Build a theory
- 3 Test the theory
- 4 Reflect and integrate

Research process models: Sequential (2)

Example:

Sharp et al (2002):

- 1 Identify the broad area of study
- 2 Select a research topic
- 3 Decide on an approach
- 4 Plan how you will perform the research
- 5 Gather data and information
- 6 Analyse and interpret these data
- 7 Present the result and findings

Research process models: Sequential (3)

Greenfield (1996):

- 1 Review the field
- 2 Build a theory
- 3 Test the theory
- 4 Reflect and integrate

Sharp et al (2002):

- 1 Identify the broad area of study
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- 4 Plan how you will perform the research
- 5 Gather data and information
- 6 Analyse and interpret these data
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What do you think about this research process model?

What is wrong with it?

(7 minutes group discussion)

Research process models: Sequential (4)

Greenfield (1996):

- 1 Review the field
- 2 Build a theory
- 3 Test the theory
- 4 Reflect and integrate

Sharp et al (2002):

- 1 Identify the broad area of study
- 2 Select a research topic
- 3 Decide on an approach
- 4 Plan how you will perform the research
- 5 Gather data and information
- 6 Analyse and interpret these data
- 7 Present the result and findings

Problems with the sequential (and generalised) process model:

- 1 Stages not subject specific
- 2 No repetition or cycles
- 3 Starting point and order fixed

Research process models: Generalised (1)

- The **generalised research process model** recognises that the stages of the research process depend on the **subject** and **nature** of the research undertaken

Example:

Data gathering and **data analysis** play no role for research in **pure mathematics** and large parts of **computer science**

Instead researchers make **conjectures** which they **prove mathematically**

- The **generalised research process model** provides **alternative routes** depending on the **subject** and **nature** of the research undertaken
- But each **route** is still **sequential**

Research process models: Generalised (2)

Example:

- (1) Identify the broad area of study
- (2) Select a research topic

In natural sciences:

- (3) Decide on an approach
- (4) Plan the research
- (5) Gather data and information
- (6) Analyse and interpret these data

In mathematics:

- (3') Make a conjecture
- (4') Prove the conjecture

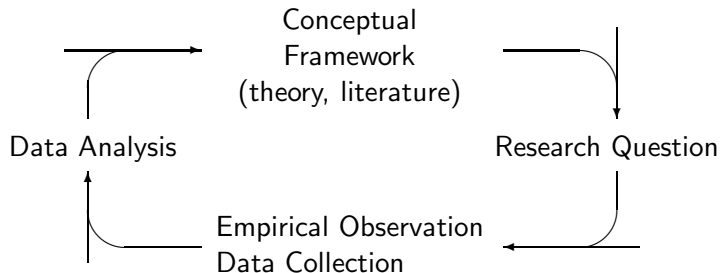
- (7) Present the result and findings

Problems with the generalised process model:

- ❶ No repetition or cycles
- ❷ Starting point and order fixed

Research process models: Circulatory

- The **circulatory research process model** recognises that any research is part of a **continuous cycle** of **discovery** and **investigation** that never ends
- It allows the research process to be **joined** at any point
- One can also **revisit** (go back to) **earlier stages**



Research process models: Evolutionary (1)

- The **evolutionary research process model** recognises that research (methods) itself **evolve** and **change over time**

That is, over time our concept of

- What research questions are admissible
- What extend and methods of data collection are possible, necessary, ethical, or reliable
- What methods are data analysis are available
- What constitutes sufficient evidence for a hypothesis
- What we mean by a systematic approach to research changes

Research process models: Evolutionary (2)

- The **evolutionary research process model** recognises that research (methods) itself **evolve** and **change over time**
- As an example, we can consider research in **mathematics**, in particular, its use of **computers**
- With respect to **mathematical proofs** we can make the following distinctions:
 - (1) Proofs created solely by humans
 - ~ typically 'sketchy', omitting steps that are considered 'obvious'
 - (2) Computer-aided mathematical proofs
 - ~ Structure and deductive steps still provided by humans, but certain computations are delegated to a computer
 - (3) Fully formal, computer generated and validated proofs
 - ~ Every step of a proof is conducted and validated by a computer, possibly under guidance by humans

Research process models: Evolutionary (3)

- The **evolutionary research process model** recognises that research (methods) itself **evolve** and **change over time**

Computer-aided mathematical proofs (1)

Four colour theorem

Any planar map can be coloured with at most four colours in a way that no two regions with the same colour share a border.

Conjectured in 1852 by Guthrie. Proved in 1976 by Appel and Haken. Proof involves a case analysis of about 10,000 cases for which the help of a computer was used

Proof seems generally accepted, but not by all Mathematician

Research process models: Evolutionary (4)

- The **evolutionary research process model** recognises that research (methods) itself **evolve** and **change over time**

Computer-aided mathematical proofs (2)

Sphere packing theorem

Close packing is the densest possible sphere packing.

Conjectured in 1611 by Kepler. Hayes published a proof plan in (1997). Execution of the plan involved solving about 100,000 linear optimisation problems using a computer. The computer files for the related programs and data requires more than 3GB of space

At one point it was suggested that the proof will be published with a disclaimer, saying that it is impossible for a human to check its correctness

Research process models: Conclusion

- Among the four common views of the **research process**
 - Sequential
 - Generalised
 - Circulatory
 - Evolutionary

the **evolutionary research process model** best describes the 'real' research process

- While the **evolutionary research process model** allows for the 'rules of the game' to change over time, this does not imply there aren't any rules
- For a young researcher it is best to follow the current established research process

Research Methods in Computer Science

Lecture 5: Intellectual discovery

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Previously ...

10 Research process models

- Sequential
- Generalised
- Circulatory
- Evolutionary

Topics

11 Scientific method

- Elements

12 Intellectual discovery

- Deduction
- Abduction
- Induction
- Process model

13 Problem solving

Scientific method

- Scientists use **observations** and **reasoning** to **develop technologies** and **propose explanations** for natural phenomena in the form of **hypotheses**
- **Predictions** from these **hypotheses** are tested by **experiment** and further technologies developed
- Any **hypothesis** which is cogent enough to make predictions can then be tested reproducibly in this way
- Once it has been established that a **hypothesis** is **sound**, it becomes a **theory**.
- Sometimes **scientific development** takes place differently with a **theory** first being developed gaining support on the basis of its logic and principles

Elements of a scientific method

The essential elements of a scientific method are [iterations](#), [recursions](#), [interleavings](#) and [orderings](#) of the following:

- [Characterisations](#)
(Quantifications, observations and measurements)
- [Hypotheses](#)
(theoretical, hypothetical explanations of observations and measurements)
- [Predictions](#)
(reasoning including logical [deduction](#) from hypotheses and theories)
- [Experiments](#)
(tests of all of the above)

Both [characterisations](#) and [experiments](#) involve data collection

Intellectual discovery

- Knowing what the **elements** of a **scientific method** are does not tell us how to come up with the right **instances** of these elements
 - What predictions does a theory make?
 - What is the right hypothesis in a particular situation?
 - What is the right experiment to conduct?
- These are commonly derived by a process involving
 - Deductive reasoning
 - Abductive reasoning
 - Inductive reasoning

Classification by Charles Sanders Peirce (1839-1914)

See <http://plato.stanford.edu/entries/peirce/> for additional details

Intellectual discovery: Deduction (1)

- **Deductive reasoning** proceeds from our knowledge of the world (theories) and predicts 'likely' observations

Example:

- Assume we know that A implies B.
- A has been observed.
- Then we should also observe B.
- Useful for **experiment generation** for theories

Example:

Newton's theory of gravity versus Einstein's theory of relativity

- Largely make the same predictions
- Both predict that the sun's gravity should bend rays of light
- However, Einstein's theory predicts a greater deflection
- Correctness of Einstein's prediction confirmed by observation in 1919

Intellectual discovery: Deduction (2)

- **Deductive reasoning** is often said **not** to lead to new knowledge
(Note: This implies pure mathematicians largely waste their time)
 - ↪ Seriously underestimates the computational effort involved in **deductive reasoning**
 - ↪ Most theories are **undecidable**
(There is no algorithm that even given infinite time could determine whether a statements follows from a theory or not)
 - ↪ Thus, establishing that a statement follows from a theory **extends** our knowledge

Intellectual discovery: Abduction

- **Abductive reasoning** proceeds from observations to causes

Example:

- The phenomenon X is observed.
- Among hypotheses A, B, C, and D,
only A and B are capable of explaining X.
- Hence, there is a reason to assume that A or B holds.

↪ Requires a **theory** linking A, B, C, D to X

- Useful for **hypothesis generation**
- **Hypotheses** must then be confirmed / eliminated through further **observation**
- It is not easy from the outside to decide whether someone uses **deduction** or **abduction**
↪ The two are often confused

Intellectual discovery: Induction (1)

- Inductive reasoning proceeds from a set of observations to a general conclusion

Example:

- Tycho Brahe, a 16th century astronomer, collected data on the movement of the Mars.
- Johannes Kepler analysed that data which was consistent with Mars moving in an elliptic orbit around the sun.
- Inductive conclusion:
Mars, and all other planets, move in elliptic orbits around the Sun, with the Sun at one of the focal points of the ellipse.
- Primary tool for theory formation

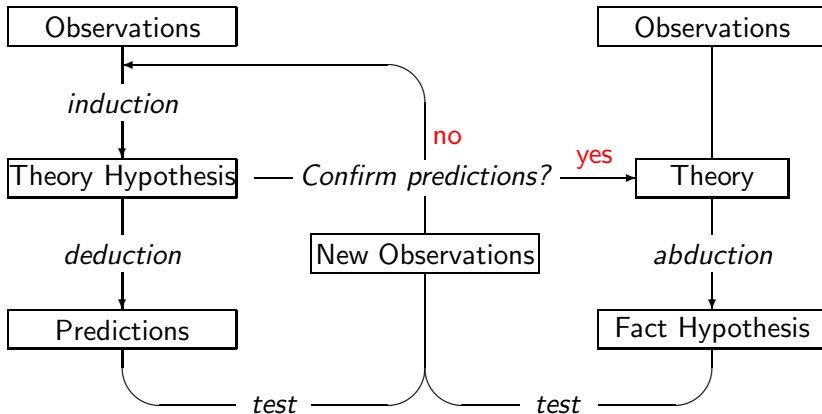
Intellectual discovery: Induction (2)

- An incomplete set of observations can easily lead to incorrect **inductive conclusions**

Example:

- All swans I've ever seen are white
- Inductive conclusion: All swans are white

Scientific method: A model



Intellectual discovery: Problems

- **Deductive reasoning** tells us that from ' A ' and ' A implies B ' we can conclude ' B '
However, it cannot tell us whether ' A ' or ' A implies B ' holds, nor whether ' B ' is what we want to show
- **Abductive reasoning** tells us that from ' B ' and ' A implies B ' we may conclude ' A '
However, it cannot tell us whether ' B ' or ' A implies B ' hold, nor how to establish that ' A ' is the case
- **Inductive reasoning** tells us that from ' $A(o_1)$ ', ..., ' $A(o_n)$ ' and ' $B(o_1)$ ', ..., ' $B(o_n)$ ' we may conclude ' $\forall x. A(x) \Rightarrow B(x)$ '.
However, it cannot tell us what the properties ' $A(-)$ ' and ' $B(-)$ ' are (nor how large the number n needs to be)

To overcome these problems we need additional techniques.

Problem solving

- **Analogy**: Look for similarity between one problem and another one already solved
- **Partition**: Break the problem into smaller sub problems which are easier to solve
- **Random/Motivated Guesses**: Guess a solution to the problem then prove it correct
- **Generalise**: Take the essential features of the specific problem and pose a more general problem
- **Particularise**: Look for a special case with a narrower set of restriction than the more general case
- **Subtract**: Drop some of the complicating features of the original problem
- **Add**: A difficult problem may be resolved by adding an auxiliary problem

Research Methods in Computer Science

Lecture 6: Research methods

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Previously ...

13 Scientific method

- Elements

14 Intellectual discovery

- Deduction
- Abduction
- Induction
- Process model

15 Problem solving

Topics

16 Classifying research

17 Research methods

- Overview
- Experiments
- Questionnaires

Classifying research (1)

Research can be classified from **three different perspectives**:

1 Field

Position of the research within a **hierarchy of topics**

Example:

Artificial Intelligence → Automated Reasoning →
First-Order Reasoning → Decidability

2 Approach

Research methods that are employed as part of the research process

Examples:

Case study, Experiment, Survey, Proof

3 Nature

- Pure theoretical development
- Review of pure theory and evaluation of its applicability
- Applied research

Classifying research (2)

- **Pure theory:**

Developing theories and working on their consequences, with regard to experimentation or application

- **Descriptive studies:**

Reviewing and evaluating existing theories, including describing the state of the art, comparing predictions with experimental data

- **Exploratory studies:**

Investigating an 'entirely' new area of research, exploring a situation or a problem

See <http://www2.uiah.fi/projects/metodi/177.htm>

- **Explanatory studies:**

Explaining or clarifying some phenomena or identifying the relationship between things

Classifying research (2)

- **Causal studies:**
Assessing the causal relationship between things
- **Normative studies:**
Producing a theory of design (or of other development) like recommendations, rules, standards, algorithms, advices or other tools for improving the object of study
- **Problem-solving studies:**
Resolving a problem with a novel solution and/or improving something in one way or another
- **Development and Application studies:**
Developing or constructing something novel

Quantitative and qualitative research methods

• Quantitative research methods

- Methods associated with **measurements** (on numeric scales)
- Stemming from natural sciences
- Used to **test hypotheses** or create a **set of observations** for inductive reasoning
- Accuracy and repeatability of vital importance

• Qualitative research methods

- Methods involving case studies and surveys
- Stemming from social sciences
- Concerned with increasing understanding of an are, rather than an explanation
- Repeatability usually a problem

Research methods (1)

• Action research:

- Pursues action (or change) and understanding at the same time
- Continuously alternates between action and critical reflection, while refining methods, data and interpretation in the light of the understanding developed in the earlier cycles

Example: Reflective teaching

• Case study:

- In-depth exploration of a single situation
- Usually generates a large amount of (subjective) data
- Should not merely report the data obtained or behaviour observed but attempt to generalise from the specific details of the situation observed

Example: Case study of open source software development

Research methods (2)

• Survey:

- Usually undertaken using questionnaires or interviews
- Questionnaire and interview design important!
(See Dawson 2005 for details)
- Determination of sample size and sample elements important!
(See specialist literature for details)

Example: Survey on the popularity or use of programming languages

• Experiment:

- Investigation of causal relationships using test controlled by the researcher
- Usually performed in development, evaluation and problem solving projects

Example: Evaluation of processor performance

Key elements of an experiment

- A precise **hypothesis** that the experiment will confirm or refute
- A completely specified **experimental system**, which will be modified in some systematic way to elicit the effects predicted by the hypothesis
- Quantitative **measurement** of the results of modifying the experimental system
- Use of **controls** to ensure that the experiment really tests the hypothesis
- **Analysis** of the measured data to determine whether they are consistent with the hypothesis
- **Report** of procedures and results so that others can replicate the experiment

Key issues for questionnaires

Consider the following questions

- What are the key issues for conducting a survey by questionnaire?
- Regarding the questionnaire itself, what types of questions do you know and what is each of them used for?

(7 minutes group discussion)

Key issues for questionnaires

- Determining the **target audience**
- Determining the most appropriate **medium**
- Achieving an acceptable **response rate**
- Ensuring **anonymity** if necessary
- Obtaining additional information about the **respondents**
- Questionnaire design
 - Layout and size (not too long, uncluttered)
 - Question types
 - (1) Quantity or information
How many hours ...
 - (2) Classification
Gender
 - (3) List or multiple choice
How do you keep informed?
 - (4) Scale
How easy is ...
 - (5) Ranking
Rank in order of importance
 - (6) Complex grid or table
Multiple classifications
 - (7) Open-ended
What do you think about ...

Research Methods in Computer Science

Lecture 7: Who is Who in Computer Science Research

Ullrich Hustadt

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University of Liverpool

Prizes and Awards

- Scientific achievement is often recognised by prizes and awards
- Conferences often give a best paper award, sometimes also a best student paper award

Example:

IJCAR Best Paper Prize

For the best paper, as judged by the program committee

- Professional organisations also give award based on varying criteria

Example:

British Computer Society Roger Needham Award

Made annually for a distinguished research contribution in computer science by a UK based researcher within ten years of their PhD.

- Arguably, the most prestigious award in Computer Science is the A. M. Turing Award

Alan M. Turing (1912-1954)

- Considered to be the father of modern computer science
- In a 1936 paper introduced **Turing machines**, as a thought experiment about the limits of mechanical computation
Gives rise to the concept of **Turing completeness** and **Turing reducibility**
- In 1939/40, Turing designed an electromechanical machine which helped to break the german Enigma code
His main contribution was an **cryptanalytic machine** which used logic-based techniques
- In the 1950 paper 'Computing machinery and intelligence' Turing introduced an experiment, now called the **Turing test**, to define a standard for a machine to be called sentient



Turing Award

The [A. M. Turing Award](#) is given annually by the Association for Computing Machinery to

an individual selected for contributions of a technical nature made to the computing community. The contributions should be of lasting and major technical importance to the computer field.

Turing Award Winners

What contribution have the following people made?

Who among them has received the Turing Award?

Frances E. Allen

Leonard M. Adleman

Paul Baran

Timothy J. Berners-Lee

Vinton G. Cerf

Edgar F. Codd

Stephen A. Cook

Lawrence J. Ellison

Douglas Engelbart

William H. Gates III

James A. Gosling

Irene Greif

Alan Kay

Donald E. Knuth

Robin Milner

Theodor H. Nelson

Lawrence Page

Alan J. Perlis

Amir Pnueli

Dennis M. Ritchie

Ronald R. Rivest

Adi Shamir

Richard M. Stallman

Ken Thompson

(12 minutes group discussion)

Turing Award Winners

What contribution have the following people made?
Who among them has received the Turing Award?

Frances E. Allen
Leonard M. Adleman
Paul Baran
Timothy J. Berners-Lee
Vinton G. Cerf
Edgar F. Codd
Stephen A. Cook
Lawrence J. Ellison
Douglas Engelbart
William H. Gates III
James A. Gosling
Irene Greif

Alan Kay
Donald E. Knuth
Robin Milner
Theodor H. Nelson
Lawrence Page
Alan J. Perlis
Amir Pnueli
Dennis M. Ritchie
Ronald R. Rivest
Adi Shamir
Richard M. Stallman
Ken Thompson

Turing Award Winners

What contribution have the following people made?

Who among them has received the Turing Award?

Frances E. Allen	✓	Alan Kay	✓
Leonard M. Adleman	✓	Donald E. Knuth	✓
Paul Baran	✗	Robin Milner	✓
Timothy J. Berners-Lee	✗	Theodor H. Nelson	✗
Vinton G. Cerf	✓	Lawrence Page	✗
Edgar F. Codd	✓	Alan J. Perlis	✓
Stephen A. Cook	✓	Amir Pnueli	✓
Lawrence J. Ellison	✗	Dennis M. Ritchie	✓
Douglas Engelbart	✓	Ronald R. Rivest	✓
William H. Gates III	✗	Adi Shamir	✓
James A. Gosling	✗	Richard M. Stallman	✗
Irene Greif	✗	Ken Thompson	✓

The ones who haven't made it (yet)

Paul Baran

One of the three inventors of packet-switched networks, along with Donald Davies and Leonard Kleinrock in the early 1960's

Timothy J. Berners-Lee

Together with Robert Cailliau invented the World Wide Web in 1989

Lawrence J. Ellison

Co-founder and CEO of the database software company Oracle

William H. Gates III

Co-founder, together with Paul Allen, of Microsoft; held the positions of CEO, chief software architect, and chairman

James A. Gosling

Invented the [Java programming language](#) in 1994; devised the original design of Java and implemented its original compiler and virtual machine

The ones who haven't made it (yet)

Irene Greif

Together with Paul Cashman introduced the concept of Computer Supported Collaborative Work (CSCW) in 1984

Theodor H. Nelson

Coined the term 'hypertext' in 1963 and published it in 1965; the idea itself goes back to 1945, Vannevar Bush's Memex device

Lawrence Page

Co-founder, together with Sergey Brin, of Google; developed the PageRank algorithm in 1998 on which Google is based

Richard M. Stallman

Software freedom activist, hacker, software developer; launched the GNU Project in 1983

- Received the Turing award in 2006

“For pioneering contributions to the theory and practice of optimizing compiler techniques that laid the foundation for modern optimizing compilers and automatic parallel execution.”



- First woman to receive the award
- Her 1966 paper on ‘Program Optimization’ and a 1971 paper with John Cocke provide the conceptual basis for the systematic analysis and transformation of computer programs
- Work forms the basis for modern machine- and language-independent **program optimizers**
- Lead an IBM project which developed the concept of **program dependence graph**, the primary structuring method used by most **parallelizing compilers** today



- Received the Turing award in 2004
"For pioneering work on internetworking, including the design and implementation of the Internet's basic communications protocols, TCP/IP, and for inspired leadership in networking."
- Led the design and implementation of the Transmission Control Protocol and Internet Protocol (TCP/IP)
- Basis for current internetworking

- Received the Turing award in 2003

*"For pioneering many of the ideas at the root of contemporary object-oriented programming languages, leading the team that developed **Smalltalk**, and for fundamental contributions to personal computing."*



- Development started in 1969, publicly available since 1980
- First complete dynamic **object-oriented programming**
- Influenced the design of **C++** and **Java**
- Included a complete **visual programming environment**
- Envisaged to be part of a 'user-centered' approach to computing



- Received the Turing award in 2002
"For their ingenious contribution for making public-key cryptography useful in practice."
- Created the world's most widely used **public-key cryptography system, RSA**, in 1977
- Clifford Cocks described an equivalent system in an internal GCHQ document in 1973, but it was never deployed and kept secret until 1997

Douglas Engelbart

- Received the Turing award in 1997

"For an inspiring vision of the future of interactive computing and the invention of key technologies to help realize this vision."



- Vision of a computer and communications based working environment
- Invention of key tools and systems that helped start the **personal computer** revolution:
 - Computer mouse
 - Multiple on screen windows
 - Linked hypermedia
 - Shared screen teleconferencing and computer aided meetings
 - Online publishing

Presented in 1968 as part of the 'mother of all demos'

- Received the Turing award in 1996

"For seminal work introducing temporal logic into computing science and for outstanding contributions to program and system verification."



- Major breakthrough in the **verification and certification of concurrent and reactive systems**
- Landmark 1977 paper 'The Temporal Logic of Programs' in Proc. 18th IEEE Symp. Found. of Comp. Sci., 1977, pp. 46–57.
 - Focus on ongoing behaviour of programs (rather than input/output behaviour)
 - Allows to easily specify qualitative progress properties of concurrent programs
 - Careful logic design enables automated verification of concurrent programs

- Received the Turing award in 1991

For three distinct and complete achievements:

- 1 *LCF*, the mechanization of Scott's Logic of Computable Functions, probably the first theoretically based yet practical tool for machine assisted proof construction;
- 2 *ML*, the first language to include polymorphic type inference together with a type-safe exception-handling mechanism;
- 3 *CCS*, a general theory of concurrency.

In addition, he formulated and strongly advanced full abstraction, the study of the relationship between operational and denotational semantics.



- Received the Turing award in 1983

*For their development of **generic operating systems theory** and specifically for the implementation of the **Unix** operating system.*



- Development of UNIX began in 1969
- Seminal paper published in 1973 on 'The UNIX Time-Sharing System' at the Fourth ACM Symposium on Operating Systems Principles
- UNIX was the first commercially important portable operating system
- Usable (almost without change) across a wide range of hardware from smartphones to supercomputers

- Received the Turing award in 1982

For his advancement of our understanding of the complexity of computation in a significant and profound way.



- Seminal paper 'The Complexity of Theorem Proving Procedures' presented at the 1971 ACM SIGACT Symposium on the Theory of Computing
- Laid the foundations for the theory of NP-completeness
- $P \neq NP$ is still one of the most fundamental open problems
- Starting point for complexity theory

- Received the Turing award in 1981

For his fundamental and continuing contributions to the theory and practice of database management systems.



- Developed the [relational approach](#) to [database management](#)
- Seminal paper published in 1970 on 'A Relational Model of Data for Large Shared Data Banks'
- Provided the impetus for widespread research into numerous related areas, including database languages, query subsystems, database semantics, locking and recovery, and inferential subsystems
- Other contributions: Boyce-Codd Normal Form
Online analytical processing (OLAP)

- Received the Turing award in 1974

For his major contributions to the analysis of algorithms and the design of programming languages.



- Author of the multi-volume book series 'The Art of Computer Programming'
 - First volume published in 1968, seven volumes planned, currently working on fourth volume
 - One of the most highly respected references in the computer science field
- Created the field of rigorous [analysis of algorithms](#)
- Creator of the [T_EX typesetting system](#) and of the Metafont font design system

- First recipient of the Turing award in 1966
For his influence in the area of advanced programming techniques and compiler construction.
- One of the developers of the ALGOL programming language



How to become a Turing award winner

To increase your chances to become a Turing award winner it might be advantageous to work in one of the following fields:

- **programming language design and implementation** (Backus, Floyd, Hoare, Milner, Naur, Iverson (APL), Dijkstra, Naur, Perlis (Algol), Fortran (Backus), Pascal, Modula (Wirth), Dahl, Nygaard (Simula), Kay (Smalltalk))
- **program compilation** (Cocke, Perlis), **program optimisation** (Allen)
- **program verification** (Floyd, Pnueli)
- analysis and theory of **algorithms** including **complexity theory** (Blum, Cook, Hopcroft, Hartmanis, Knuth, Karp, Rabin, Scott, Stearns, Tarjan, Yao)
- theory and practice of **databases** (Bachman, Codd, Gray)
- theory and practice of **operating systems** (Brooks, Corbató, Ritchie, Thompson, Lampson)
- **artificial intelligence** (Feigenbaum, Minsky, Newell, Reddy, Simon)

Research Methods in Computer Science

Lecture 8: Reading research paper

Ullrich Hustadt

Department of Computer Science
University of Liverpool

Topics

18 Practical 1

19 Reading research papers

Today's questions

- 1 Each of you has compiled a list of five to ten concepts that you did not understand. Discuss those, see whether someone else in your group can give you an explanation, then, as a group, compile a short list of concepts that remain unclear
- 2 For each of the papers list at least three claims that they put forward and note what evidence they provide to support those claims

(15 minutes group discussion)

Practical 1: Claims and Evidence

Paper 1: 'A SAT-based decision procedure for $\mathcal{ALC}$ '

- ① KSAT outperforms KRIS by several orders of magnitude
↪ empirical evidence on randomly generated samples
- ② SAT-based decision procedures are intrinsically bound to be more efficient than tableau-based decision procedures
↪ in contrast to SAT-based procedures, tableau-based procedures consider the same truth assignment more than once
- ③ There is partial evidence of an easy-hard-easy pattern
↪ empirical evidence; easy-hard-easy pattern evident for KSAT (but not KRIS)

Practical 1: Claims and Evidence

Paper 2: 'On evaluating decision procedures for modal logic'

- ① KSAT does not qualitatively outperform *KRIS* with pre-processing
 - ↔ empirical evidence on the same randomly generated samples, but using *KRIS* with pre-processing
- ② Non-eager application of simplifications causes inferior performance of *KRIS* (even with pre-processing)
 - ↔ empirical evidence and analysis of behaviour of *KRIS*
- ③ Easy-hard-easy pattern is an artificial phenomenon of KSAT
 - ↔ empirical evidence; translation approach has no easy-hard-easy pattern although it solves the hardest samples faster than KSAT

Practical 1: Claims and Evidence

Paper 3: 'More evaluation of decision procedures for modal logics'

- 1 All the claims in Paper 1 are correct
 - ← empirical evidence on a new, less flawed set of randomly generated samples; repetition of the argument about truth assignments
- 2 KSATC also outperforms the translation approach ← empirical evidence

Practical 1: Conclusion

Tableau methods construct refutations by **case distinction** and the application of **decomposition rules**

- **SAT-based procedures** can be seen as **tableau methods** using
- a specific kind of **case distinction** and
 - a specific **strategy** for the application of **decomposition rules**
- Question is not **tableau-based** vs **SAT-based** procedures, but
- what kind of **case distinction** is best (if any) and
 - what **strategy** for the application of **decomposition rules** is best (if any)

These questions are still open

- sets of modal logic/description logic expressions are infinite
- lack of 'real-world' samples

Practical 1: Learning points

Regarding **research papers**:

- Notions need precise definitions
- Claims need to be formulated unambiguously
- Evidence needs to be constructed carefully

Regarding the **research process**:

- Peer review does not prevent mistakes
- Research can progress without 'resolving contradictions'
- Research is also a social process

Reading research papers

- Research aims to add the world's body of knowledge
 ~> Requires a researcher to be aware of what the world's body of knowledge (in the area s/he works in)
- Frontiers of the world's body of knowledge are **not** documented in text books, but in

	journal articles	
reliability ↑	conference papers	
	workshop papers	↓ timeliness
	technical reports	

Get organised

- Maintain a database of all the books and papers you read
- Data stored should at least include title, author, place of publication, and storage location
- Preferably you should also keep a record of the answers to some or all of the following questions:
 - 1 What is the main topic of the article?
 - 2 What was/were the main issue(s) the author said they want to discuss?
 - 3 Why did the author claim it was important?
 - 4 How does the work build on other's work, in the author's opinion?
 - 5 What simplifying assumptions does the author claim to be making?

Get organised

- Maintain a database of all the books and papers you read
- Data stored should at least include title, author, place of publication, and storage location
- Preferably you should also keep a record of the answers to some or all of the following questions:
 - 6 What did the author do?
 - 7 How did the author claim they were going to evaluate their work and compare it to others?
 - 8 What did the author say were the limitations of their research?
 - 9 What did the author say were the important directions for future research?

Evaluating research papers

- Whenever you read a research paper, you should try to **evaluate** at the same time.
- Try to **answer the following questions**:
 - 1 Is the topic of the paper sufficiently interesting (for you personally or in general)?
 - 2 Did the author miss important earlier work?
 - 3 Are the evaluation methods adequate?
 - 4 Are the theorems and proofs correct?
 - 5 Are arguments convincing?
 - 6 Does the author mention directions for future research that interest you?
- Given the answers to these questions for a number of research papers, you should be able to construct a **research proposal** by considering how you could improve the work presented in them

Research Methods in Computer Science

Lecture 9: Structure of research papers

Ullrich Hustadt

Department of Computer Science
University of Liverpool

Previously ...

18 Practical 1

19 Reading research papers

Topics

20 Structure of research papers

21 Hints

Today's questions

Taking the research papers you have been given and others that you may have come across in the past as a (hopefully) representative sample, consider the following questions:

- 1 What elements constitute the structure of the papers? Are the elements and their order identical for all the papers? If not, which elements do the papers have in common and which elements only appear in only some of the papers?
- 2 What characterises each of the elements of the papers? That is, looking at each element of each of the papers, what do they have common?

(15 minutes group discussion)

Structure of a research paper

- ➊ Title
- ➋ List of authors (and their contact details)
- ➌ Abstract
- ➍ Introduction
- ➎ Related Work (either part of or following introduction or before summary).
- ➏ Outline of the rest of the paper
- ➐ Body of the paper
- ➑ Summary and Future Work (often repeats the main result)
- ➒ Acknowledgements
- ➓ List of references

Title

- As short as possible, but without abbreviations or acronyms (unless they are commonly understood)
- As specific as necessary and as general as possible
(e.g. 'The Complexity of Theorem-Proving Procedures'
 $\leadsto$ introduced the notion of 'NP-Completeness'
 $\leadsto$ starting point of complexity theory)
- Include key phrases which are likely to be used in a search on the topic of the paper
(e.g. 'modal logic', 'calculus', 'decision procedure')
- Avoid phrases which are too common
(e.g. 'novel')
- Use phrases that describe distinctive features of the work
(e.g. 'Real-world Reasoning with OWL')

Authors (1)

- An **author** of a paper is an individual who
 - ① made a significant **intellectual contribution** to the work described in the paper
(in contrast, for example, to a **monetary contribution**);
 - ② made a contribution to **drafting, reviewing and/or revising** the paper for its **intellectual contribution**
(in contrast, for example, to **spell checking** or **typesetting**); and
 - ③ approved the final version of the paper including references

Some organisations / publishers have strict rules regarding **authorship**

- Order of **authors** may depend on
 - **subject area**: pure theory $\leadsto$ often **alphabetical**
applied research $\leadsto$ often based on **contribution**
 - **research assessment**
(e.g. bibliographic measures associating order with contribution)
 - **cultural context**

Authors (2)

- In Computer Science, **academic degrees** and **membership of professional organisations** are typically not indicated
- List of authors is typically followed by **contact information** consisting of **affiliation** and **e-mail address** (not postal address)
- Some journals allow authors to provide longer descriptions of themselves including photographs

Abstract

- Typically not more than 100–150 words
- Should aim to **motivate** people to read the paper
- Highlight the **problem** and the **principal results**
- The abstract will be included in **literature databases**
 - ~> Make sure **key phrases** which might be used in searches are included (same principle as for titles)
- Keep **references** to a minimum
- Keep **equations** and other **mathematical expressions** to a minimum

Introduction

- State the general **area of research**
(unless this is obvious from the context in which the paper appears)
- Introduce the **problem**
state why the problem is important and/or interesting
- Outline the **approach** taken to solve the problem
- Outline the **solution** or **principal results**
state why the results are important and/or interesting
- Do not repeat the **abstract**
- Avoid platitudes and cliches

Related work

- **Related work** is previous work by the same or other authors which addresses the same or closely related **problems** / **topics**
- Section on **related work** gives **credit** to such work and **establishes** the **originality** of the current work
- Extent depends on the **space available** and **relevance of the related work** to the work presented in the paper

Within these two constraints, make sure all related work is cited and correctly described

- Failure to give credit can result in a bad evaluation and kill your paper
- Section on related work is either part of the **introduction** or is placed **at the end of the body of the paper**

Outline of the paper

- Typically at the end of the [introduction](#)
- Describes the content of the [body of the paper](#) section by section

Example:

The remainder of the paper is organised as follows. In Section 2, we introduce ... Section 3 describes ... Finally, we describe future work in Section 5.

(Note that 'Section' is capitalised.)

Body of the paper

- Depends strongly on subject area and topic of the paper
- Typical structure of a Computer Science paper on **theoretical research**:
 - 1 Basic definitions
 - 2 Description of a new algorithm, calculus, or formalism
 - 3 Sequence of **theorems** accompanied by **proof** or proof sketches
 - 4 Applications / consequences of the results (optional)
- Typical structure of a Computer Science paper on **applied research**:
 - 1 Architecture of a new system
 - 2 Description of the realisation
 - 3 Evaluation
- Combinations of the two are possible and quite typical
- Papers on **action research**, **case studies**, **surveys**, **experiments** are also common and have their own structure

Conclusion and/or Future Work

- Summarises the **contributions** of the paper
- Describes the **implications** and/or **applications** of the **contributions** made by the paper
- Outlines **future directions** of **research**

Acknowledgements

- Acknowledges external **funding sources**
- Thanks **non-authors** that made a significant contribution
 - **colleagues** or **fellow researchers** with which the authors had discussions related to the topic of the paper
 - **anonymous referees** provided they have given exceptional level of feedback or important insights

List of references

See lectures on citing and referencing

Hints

Top-down design: Start with an outline, then fill in the details

Inside-out writing: Fill in the body of the paper first, then write introduction, related work, conclusion; finally, write the abstract

Diagrams/Tables: Are all diagrams and tables readable? Can they be understood?

Dependency analysis: Is the paper self-contained and are notions presented in the correct order?

Factuality: Make sure everything stated in the paper is factually correct

Interpretability: For each sentence check whether it could be misread; if so, try to fix it

Optimisation: Remove unnecessary parts, shorten exposition

Readability: Does it read well? Are all parts interconnected?

Additional guidance

- Alan Bundy. How to Write an Informatics Paper.
<http://tinylink.com/?epHuLuq60m>.
(Accessed 3 October 2007).
- Simon Payton Jones. How to write a great research paper.
<http://tinylink.com/?vGPkhu7VeA>.
(Accessed 3 October 2007).
- Jennifer Widom. Tips for Writing Technical Papers.
<http://infolab.stanford.edu/~widom/paper-writing.html>.
January 2006 (accessed 3 October 2007).

Research Methods in Computer Science

Lecture 10: Literature searches

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University of Liverpool

Previously ...

20 Structure of research papers

21 Hints

Topics

22 Literature sources

23 Databases and search engines

- Publishers
- Literature DBs
- Web search engines
- Comparison

24 Queries

Searching for literature

- What are you trying to find out?
 - ~> Try to specify exactly what you need to know
- What type of information do you want to find?
 - ~> An answer to a specific question?
 - ~> An overview of a subject area?
 - ~> A specific document?
- Why do you need this information?
 - ~> Literature survey: Information needs to be comprehensive
 - ~> Short essay: Limited number of sources is sufficient
- How quickly do you need the information?
 - ~> Immediately: Internet
 - ~> In a day: Library
 - ~> In a week: Inter Library Loans

Searching for literature

Consider the following tasks:

- 1 Obtain a paper copy of the following article:

*P. McBurney, S. Parsons and M. Wooldridge (2002):
Desiderata for agent argumentation protocols.
In: C. Castelfranchi and W. L. Johnson (Editors):
Proceedings of the First International Joint Conference on
Autonomous Agents and Multi-Agent Systems (AAMAS
2002), pp. 402–409, Bologna, Italy. July 2002. New York,
USA: ACM Press.*

- 2 Find out which other publications refer to the article above.

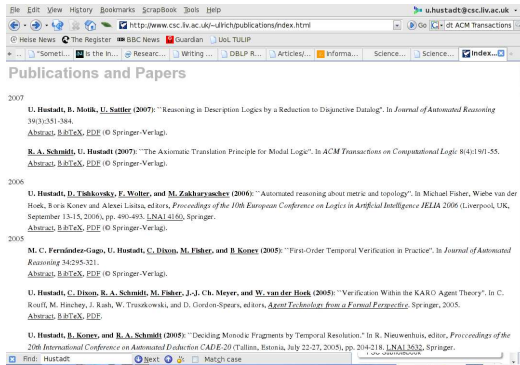
How would you accomplish these tasks?

(7 minutes group discussion)

Where to search: Sources

Sources for literature on the internet:

- Freely available collections (personal/institutional)



- Publishers' websites/databases
- Literature databases

Where to search: Sources

Sources for literature on the internet:

- Freely available collections (personal/institutional)
- Publishers' websites/databases

The screenshot shows a web browser window displaying a SpringerLink article. The article title is "Reasoning in Description Logics by a Reduction to Disjunctive Datalog" by Ulrich Hustadt, Boris Motik, and Ulrike Sattler. The article is from the "Journal of Automated Reasoning". A prominent orange banner states "Read for FREE: the #1 cited article of 2005 from UCV". The abstract discusses applications of description logics to efficient reasoning with knowledge bases. The page includes a sidebar with SpringerLink navigation options and a right sidebar with search and export options.

- Literature databases

Where to search: Sources

Sources for literature on the internet:

- Freely available collections (personal/institutional)
- Publishers' websites/databases
- Literature databases

The screenshot shows a web browser window with the address bar displaying a Scopus search result URL. The search results are displayed in a table format with columns for Document (sort by relevance), Author(s), Date, Source Title, and Cited By. The results list six papers related to modal logic and reasoning.

Document (sort by relevance)	Author(s)	Date	Source Title	Cited By
1. Reasoning in description logics by a reduction to disjunctive datalog Abstract - Refs Full Text Full Text Full Text	Hustadt, U., Motik, B., Sattler, U.	2007	Journal of Automated Reasoning 39 (3), pp. 351-384	0
2. The axiomatic translation principle for modal logic Abstract - Refs Full Text Full Text Full Text	Schmidt, R.A., Hustadt, U.	2007	ACM Transactions on Computational Logic 8 (4), art. no. 1276921	0
3. Automated reasoning about metric and topology Abstract - Refs Full Text Full Text Full Text	Hustadt, U., Tishkovsky, D., Walter, F., Zakharenchev, M.	2006	Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 4160 LNAI, pp. 490-493	0
4. Deciding monodic fragments by temporal resolution Abstract - Refs Full Text Full Text Full Text	Hustadt, U., Konev, B., Schmidt, R.A.	2005	Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 3632 LNAI, pp. 204-218	1
5. A decomposition rule for decision procedures by resolution-based calculi Abstract - Refs Full Text Full Text Full Text	Hustadt, U., Motik, B., Sattler, U.	2005	Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 3452 LNAI, pp. 21-35	0
6. Mechanising first-order temporal resolution Abstract - Refs Full Text Full Text Full Text	Konev, B., Degtyarev, A., Dixon, C., Fisher, M., Hustadt, U.	2005	Information and Computation 199 (1-2), pp. 55-86	1

Where to search: Interrelationship of Sources

- ① Authors submit paper to conference/journal for **peer review**
- ② If accepted, the paper is **revised** by the authors and submitted to conference/journal editor
- ③ The paper is **processed** to bring it into the publisher's format (typesetting/layout)
- ④ The paper is then
 - included in the **publisher's database**,
 - made available on-line via the **publisher's website**, and
 - possibly published in printed form
(not necessarily in that order)
- ⑤ **Literature databases**
 - collect the bibliographic information from several publishers, and
 - add additional information (references with links, citation index)
 - link back to publisher for full-text of papers

Databases and search engines: Publishers

The University Library has subscriptions to many publishers' databases:

ACM Digital Library	Full-text of all ACM journals and conference proceedings http://portal.acm.org.ezproxy.liv.ac.uk/dl.cfm
IEEE Xplore	Full-text of IEEE journals, conference proceedings, and books http://ieeexplore.ieee.org.ezproxy.liv.ac.uk/
ScienceDirect	Full-text of Elsevier journals http://www.sciencedirect.com.ezproxy.liv.ac.uk
SpringerLink	Full-text of Springer journals, conference proceedings, and books http://www.springerlink.com.ezproxy.liv.ac.uk/
Wiley Inter-Science	Full-text of Wiley journals and books http://www.interscience.wiley.com.ezproxy.liv.ac.uk/

Access to full-text requires authentication by MWS login and password

Databases and search engines: Literature databases

The University Library has subscriptions to many literature databases:

Scopus	Covers 14,000 journals and proceedings series; incl. ACM, Elsevier, IEEE, Springer http://www.scopus.com/
Web of Knowledge	Covers 22,000 journals and 192,000 proceedings; incl. ACM, Elsevier, IEEE, Springer http://isiknowledge.com/
Metalib (UoL)	Meta search engine for ACM Digital Library, IEEE Explore, etc but also Scopus, Web of Science and Google Scholar http://www.liv.ac.uk/library/electron/

Adding `.ezproxy.liv.ac.uk` to the server name again allows access from outside the campus using your MWS login and password for authentication

Databases and search engines: Web search engines

Freely available (scholarly) web search engines include:

Citeseer	Digital library of 750k freely available papers in computer and information science http://citeseer.ist.psu.edu
Google	General internet search engine http://www.google.co.uk
Google Scholar	Searches scholarly literature on the web. http://scholar.google.com
Scirus	Searches journals (ScienceDirect) and web resources http://www.scirus.com/
Windows Live Search Academic	Academic search engine - search academic journals and content for article titles, author names, article abstracts, and conference proceedings. http://academic.live.com/

Databases and search engines: Comparison

- All these databases and search engines, and many more, are accessible from on central point:

http://dbweb.liv.ac.uk/library_resources/ohecampus/comp.asp

- The library's own catalogue is available at

<http://library.liv.ac.uk/>

- There is an important difference to remember:

Library catalogue: Allows to search **for** a journal, but not **for** journal articles

Publishers' and literature databases: Allow to search **for** journal articles, but not **in** the full-text journal articles

Web search engines: Allow to search **in** the full-text of journal articles, but have difficulties with their structure

Databases and search engines: Comparison

- **Literature databases** cover a vast number of journals and conferences, but
 - they do not cover all journals and conference
 - they do not cover textbook, handbooks, collections of articles in book form
 - they do not cover workshops and similar scientific meetings
 - they do not cover technical reports and pre-prints
- **Web search engines** provide much better coverage of these types of publications, but
 - typically also return a lot of irrelevant material to a query
 - leave it to the user to distinguish high quality from low quality material

Queries (1)

- Search terms might be simple keywords, phrases, or consist of field identifiers, modifiers, operators, and keywords

Examples: induction
"mathematical induction"
induct*
author = Ambuhl
author like Ambuhl
author soundex(Maier)

- Queries are typically constructed from search terms using boolean operators

Examples: induction AND mathematical
induction OR deduction
induction AND NOT recruitment

Queries (2)

- Queries are typically constructed from **search terms** using **boolean operators**
 - **AND** retrieves records where ALL of the search terms are present,
induction AND mathematical
 - **OR** retrieves records containing either one term OR another
induction OR deduction
 - **NOT** retrieves records NOT containing a particular term
NOT recruitment
- The set of all correct queries for a particular search engine is its **query language**
- Typically, **different search engines** use **different query languages**

Keywords

- Only the right **keywords** will correctly identify useful information
- **Mode** of search is very important:
 - **narrow**: you are looking for exactly one record
 - ~ use a **search term** which is as specific as possible
“cell microprocessor” instead of **cell**
 - ~ use additional criteria
 - publication date **year** = 2006
 - type **type** = journal
 - language **language** = english
 - publisher **publisher** = Springer
 - **wide**: you are looking for all records relating to a subject

Keywords

- Only the right **keywords** will correctly identify useful information
- **Mode** of search is very important:
 - **narrow**: you are looking for exactly one record
 - **wide**: you are looking for all records relating to a subject
 - ~> try alternative words/phrases
microprocessor / computer processor / computer chip
 - ~> try alternative spellings
judgement / judgment
 - ~> try **wildcards**
gene* for genes, genetics, genetically

Conducting a search

- ❶ Construct a query
- ❷ Search the databases, starting with the literature databases then moving to web search engines
- ❸ Record all useful references
 - ↪ some databases allow export in a format that can be Record imported in [RefWorks](#) or [EndNote](#) enough information for someone to be able to find it again
- ❹ After having searched two or three sources, review the progress of the search
 - too little relevant sources found so far ↪ modify query

Research Methods in Computer Science

Lecture 11: Bibliographies and Referencing (1)

Ullrich Hustadt

Department of Computer Science
University of Liverpool

Previously ...

- 22 Literature sources
- 23 Databases and search engines
 - Publishers
 - Literature DBs
 - Web search engines
 - Comparison
- 24 Queries

Today ...

- 25 Referencing
 - Introduction
 - Types of work
 - More examples

Today's questions

Discuss the following questions:

- 1 Why do we cite the work of others?
- 2 What constitutes a good source?
- 3 What information about a source should be included in a list of references?

(15 minutes group discussion)

References (1)

Why do we cite the work of others?

- 1 To acknowledge the work of other writers and researchers
- 2 To demonstrate the body of knowledge on which our own work is based
- 3 To enable the reader to trace our sources easily and lead her/him on to further information

We do NOT cite to indicate that we have copied text from another source! That's plagiarism!

Plagiarism

According to the University's definition, **plagiarism** is:

- the verbatim (word for word) copying of another's work without appropriate and correctly presented acknowledgement;
- the close paraphrasing of another's work by simply changing a few words or altering the order of presentation, without appropriate and correctly presented acknowledgement;
- unacknowledged quotation of phrases from another's work;
- the deliberate and detailed presentation of another's concept as one's own.

Copying of another's work, then adding a reference to that work, is NOT considered an 'appropriate and correctly presented acknowledgement'

Verbatim copying is only allowed in the context of **proper quotation**

References (2)

- What constitutes a good source?

- 1 Precise location

- ~> Sufficient information must be given for a third person to be able to locate your source

- 2 Longevity of source

- (Journals → Proceedings → Technical Reports → Web sources)

- 3 Accessibility of source

- ~> Completely free → Free subscription → Paid

- ~> Avoid 'private communication'

- 4 Reputation / Quality of source

- 5 'Originality'

- Original paper → secondary paper / translation

- 6 'Language'

- If possible, a source should be in the language you write in

- 7 Readability of source

- Well written → badly written

Vocabulary

Citing / Referencing

Formally recognising, within your text, the sources from which you have obtained information

Citation / Quotation

A passage or words quoted within your text, supported with a reference to its source

Reference

A detailed description of a source from which you have obtained information

List of references

List of all sources which are cited in the body of your work

Bibliography

List of all sources which have been consulted in preparation of your work

Citing: Rules of thumb (Zobel 2004)

- If you discuss a paper in detail or note some particular contribution it makes, it **must** be cited
- Claims, statements of fact, discussions of previous work **should** be supported by references, if not supported by your current work
But: Do not cite to support common knowledge;
do not end every sentence with a reference
- References to your own previous work is **allowed** if it is relevant to your current work
But: Gratuitous self-reference is counterproductive
- Attribute work correctly, in particular, when relying on secondary sources
 - Bad:** According to Dawson (1981), stable graphs have been shown to be closed
 - Good:** According to Kelly (1959; as quoted by Dawson, 1981), stable graphs are closed

References

- References need to include the following information, with the order and format depending on the chosen **style**:
 - Author(s) or editor(s) responsible for writing/editing the work cited
 - Title and subtitle of the work
 - Where the work can be obtained or found
 - Year the work was created, presented, and/or published
- What information is required about where the work can be obtained depends on its **type**

References: Types of work

Book

- Author(s) or editor(s)
- Title and subtitle
- Edition, if not the first, for example 2nd ed.
- Series and individual volume number (if any)
- Publisher
- (Place of publication)
- Year of publication

Examples:

A. A. Fraenkel, Y. Bar-Hillel, and A. Levy. *Foundations of Set Theory*, 2nd revised edition. Studies in Logic and The Foundations of Mathematics 67. North-Holland, Amsterdam, 1973.

A. Robinson and A. Voronkov, editors. *Handbook of Automated Reasoning*. Elsevier, 2001.

References: Types of work

Chapter/section of a book

- Author(s) of the chapter/section
- Title and subtitle of the chapter/section
- Author/editor of collected work
- Title and subtitle of collected work
- Chapter/section referred to
- Page numbers of chapter/section referred to
- Publisher
- (Place of publication)
- Year of publication

Example:

W. Bibel and E. Eder. Methods and calculi for deduction. In C. J. Hogger, D. M. Gabbay and J. A. Robinson, editors, *Handbook of Logic in Artificial Intelligence and Logic Programming, Volume 1*, chapter 3, pages 67–182. Oxford University Press, 1993.

References: Types of work

Conference proceedings

- Editor(s) of proceedings
- Name and number of conference
- Location of conference (if appropriate)
- Time of conference
- Title of published work; if different from the name of the conference
- Series and individual volume number (if any)
- Publisher
- Place of publication
- Year of publication

Example:

D. A. Basin and M. Rusinowitch, editors. *Automated Reasoning - Second International Joint Conference, IJCAR 2004, Cork, Ireland, July 4–8, 2004, Proceedings*. Lecture Notes in Computer Science 3097. Springer, 2004.

References: Types of work

Conference paper

- Author(s) of the paper
- Title and subtitle of the paper
- All information on the conference proceedings plus
- Page numbers of the paper

Example:

Volker Weispfenning. Solving Constraints by Elimination Methods. In D. A. Basin and M. Rusinowitch, editors. *Automated Reasoning - Second International Joint Conference, IJCAR 2004, Cork, Ireland, July 4–8, 2004, Proceedings*. Lecture Notes in Computer Science 3097, p. 336–341. Springer, 2004.

References: Types of work

Journal article

- Author(s) of the article
- Title and subtitle of the article
- Title of the journal
- Volume and part number
- Page numbers of article
- Date, month or season of the year, if appropriate
- Year of publication

Note: Information on publisher is typically not required

Examples:

R. MacGregor. Inside the LOOM description classifier. *SIGART Bulletin*, 2(3):88–92, 1991.

A. Seager. Energy subsidy plan for home runs out of cash. *The Guardian*, 21 October 2006, p. 6.

References: Types of work

Thesis and dissertation

- Author of the work
- Title and subtitle of the work
- Type of work
- Awarding institution including its address
- Year, possibly month, of publication

Examples:

G. Rosu. *Hidden Logic*. PhD thesis, Department of Computer Science and Engineering, University of California, San Diego, CA, USA, August 2000.

R. A. van der Goot. *Strategies for modal resolution*. Master's thesis, Faculty of Technical Mathematics and Informatics, Delft University of Technology, The Netherlands, 1994.

References: Types of work

Web pages

- Author(s) of the web page(s)
- Title and subtitle
- URL
- Date of last modification, if available
- Date of access

Examples:

The PHP Group. PHP: Hypertext preprocessor.

<http://www.php.net/>. 22 October 2006.

The International DOI Foundation. The Digital Object Identifier System. <http://www.doi.org/>. 25 July 2006 (accessed 22 October 2006).

More examples

Bad:

Marco Dorigo and Thomas Stutzle, Ant Colony Optimization.

Good:

Marco Dorigo and Thomas Stützle. Ant Colony Optimization. [Bradford Book, 2004.](#)

More examples

Bad:

JAVA, JAVA, JAVA by Ralph Morelli

Good:

Ralph Morelli. Java, Java, Java: Object-Oriented Problem Solving, 2nd edition. Prentice Hall, 2003.

More examples

Bad:

Marco Dorigo, Gianni Di Caro, Michael Samples, Ant Algorithms, third international workshop, Ant 2002, Brussels, Belgium, September 2002, Proceedings.

Good:

Marco Dorigo, Gianni Di Caro, [and](#) Michael Samples, [editors](#). Ant Algorithms: Third International Workshop, ANTS 2002, Brussels, Belgium, September [12–14](#), 2002, Proceedings. [Lecture Notes in Computer Science 2463](#). Springer, 2002.

More examples

Bad:

`http://www.cut-the-knot.org/blue/Stern.shtml`

Good:

Alexander Bogomolny. Stern-Brocot Tree.

`http://www.cut-the-knot.org/blue/Stern.shtml`.

Last modification June 17, 2000. Accessed October 26, 2006.

Research Methods in Computer Science

Lecture 12: Bibliographies and Referencing (2)

Ullrich Hustadt

Department of Computer Science
University of Liverpool

Previously ...

- 25 Referencing
 - Introduction
 - Types of work
 - More examples

Today ...

26 Bibliography styles

- Ordinal number
- Author-date
- Abbreviation

27 Using EndNote for Citations and Bibliographies

- Introduction
- References
- Word
- Conclusion

Styles: Ordinal number

- Sources listed in the bibliography are sorted according to some ordering, typically based on the authors' names, and numbered consecutively
- References in the text are given as (lists of) numbers cross-referencing the bibliography, enclosed in square brackets

Example:

Key techniques for utilising temporal logic specifications have been investigated, including verification via proof [3] and verification via model-checking [1,2].

Bibliography

1. E. Clarke, O. Grumberg, and D. A. Peled. *Model Checking*. MIT Press, 2000.
2. K. L. McMillan. *Symbolic Model Checking*. Kluwer, 1993.
3. M. Vardi and P. Wolper. Reasoning about infinite computations. *Inform. and Computat.*, 115:1–37, 1994.

Styles: Author-date (1)

- Sources in the reference list are arranged alphabetically by the authors' names;
where there is more than one work by the same authors, they are arranged by year of publication, starting with the earliest;
where there is more than one work with the same authors and date, a letter is added to the year of publication to distinguish them

Example:

Bibliography

- E. Clarke, O. Grumberg, and D. A. Peled (2000). *Model Checking*. MIT Press.
- K. L. McMillan (1993). *Symbolic Model Checking*. Kluwer.
- M. Vardi and P. Wolper (1994). Reasoning about infinite computations. *Inform. and Computat.*, 115:1–37.

Styles: Author-date (1)

- Sources in the reference list are arranged alphabetically by the authors' names;
where there is more than one work by the same authors, they are arranged by year of publication, starting with the earliest;
where there is more than one work with the same authors and date, a letter is added to the year of publication to distinguish them

Example:

Bibliography

P. Wolper (1996a). Where is the Algorithmic Support? *ACM Comput. Surv.* 28(4): 58.

P. Wolper (1996b). The Meaning of "Formal". *ACM Comput. Surv.* 28(4): 127.

Styles: Author-date (2)

- A reference is given by the authors' names and the date enclosed in parentheses unless the authors' names are part of the sentence

Example of quoting:

The following is an extract from (Wolper 1996a):

Consider, for instance, the issue of compositionality in proof systems for concurrency. I am not going to argue that compositionality is undesirable, but that achieving it without algorithmic support (in a broad sense) is easy and mostly useless.

Example of citing:

While Wolper (1996a) does not argue that compositionality in proof systems for concurrency is undesirable, he claims that achieving it without algorithmic support is mostly useless.

Styles: Author-date (2)

- A reference is given by the authors' names and the date enclosed in parentheses unless the authors' names are part of the sentence

Examples:

Recent work (Wolper 1996a, 1996b) stresses the importance of algorithmic support for formal methods.

Wolper (1996a, 1996b) stresses the importance of algorithmic support for formal methods.

The completion procedure may fail in general, but has been extended to a refutationally complete theorem prover (cf. Lankford 1975, Hsiang and Rusinowitch 1987, and Bachmair, Dershowitz and Plaisted 1989).

Completion procedures for conditional equations have been described by Kounalis and Rusinowitch (1988), and by Ganzinger (1987a, 1987b).

Styles: Abbreviation (1)

- Mix of ordinal number style and author-date style
- Sources in the bibliography are presented like in ordinal number style, but instead of numbering them, each source is given a unique identifier based on authors' names and year of publication, with additional letters to disambiguate duplicate abbreviations

Example:

Bibliography

- [CGP00] E. Clarke, O. Grumberg, and D. A. Peled. *Model Checking*. MIT Press, 2000.
- [vdG94] R. A. van der Goot. *Strategies for modal resolution*. Master's thesis, Delft University of Technology, The Netherlands, 1994.
- [Wol96a] P. Wolper. Where is the Algorithmic Support? *ACM Comput. Surv.* 28(4):58, 1996.
- [Wol96b] P. Wolper. The Meaning of "Formal". *ACM Comput. Surv.* 28(4):127, 1996.

Styles: Abbreviation (2)

- References in the text are given as (lists of) abbreviations cross-referencing the bibliography, again enclosed in square brackets

Examples:

Key techniques for utilising temporal logic specifications have been investigated, including verification via proof [VW94] and verification via model-checking [CGP00,McM93].

Recent work [Wol96a, Wol96b] stresses the importance of algorithmic support for formal methods.

Wolper in [Wol96a,Wol96b] stresses the importance of algorithmic support for formal methods.

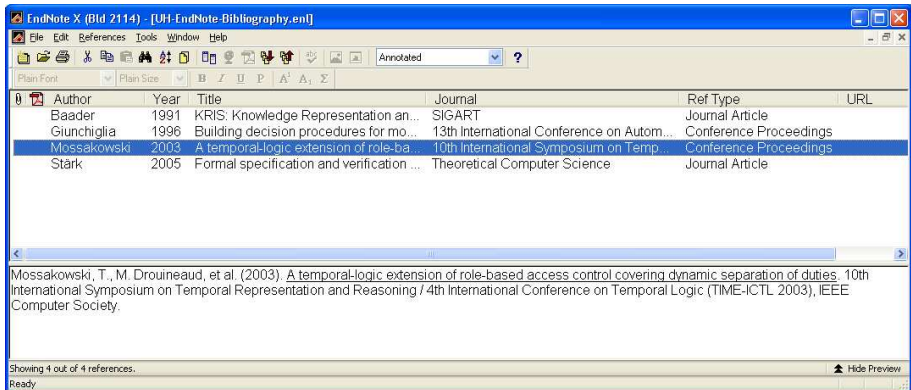
The completion procedure may fail in general, but has been extended to a refutationally complete theorem prover (cf. [Lan75,HR87,BDP89]). Completion procedures for conditional equations have been described by Kounalis and Rusinowitch [KT88], and by Ganzinger [Gan87a,Gan87b].

Organising references

- There are myriads of styles for references and bibliographies
- You should maintain information on your sources in a 'neutral' format
- Ideally, you should use a tool which
 - supports such a 'neutral' format
 - allows to add, delete, modify references
 - allows to search for references
 - interacts with your word processor/text editor
 - generates a list of references in any desired format

Organising references: EndNote

EndNote is a **reference manager** available for Microsoft Windows and Mac OS X which interfaces with Microsoft Word



EndNote: Entering references

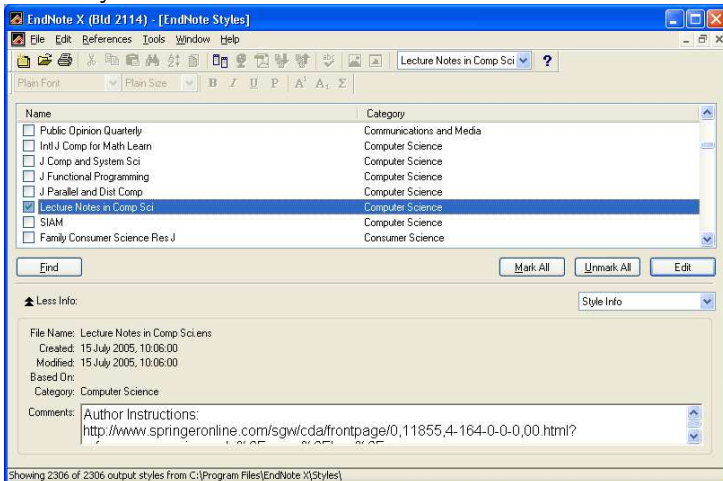
The screenshot shows the EndNote X (Bld 2114) - [Mossakowski, 2003 #3] window. The menu bar includes File, Edit, References, Tools, Window, and Help. The toolbar contains various icons for file operations and editing. The 'Reference Type' dropdown is set to 'Conference Proceedings'. The 'Hide Empty Fields' button is visible. The form fields are as follows:

Field	Value
Author	Mossakowski, Till Drouineaud, Michael Sohr, Karsten
Year of Conference	2003
Title	A temporal-logic extension of role-based access control covering dynamic separation of duties
Editor	
Conference Name	10th International Symposium on Temporal Representation and Reasoning / 4th International Conference on Temporal Logic (TIME-ICTL 2003)
Conference Location	
Publisher	IEEE Computer Society
Volume	
Number of Volumes	
Pages	83-90

The status bar at the bottom left indicates 'Ready'.

EndNote: Choosing bibliography styles (1)

EndNote allows you to format your citations and your bibliography in a number of styles



EndNote: Choosing bibliography styles (2)

EndNote X (Bld 2114) - [UH-EndNote-Bibliography.ent]

File Edit References Tools Window Help

Lecture Notes in Comp Sci ?

Plain Font Plain Size B I U P A¹ A₁ Σ

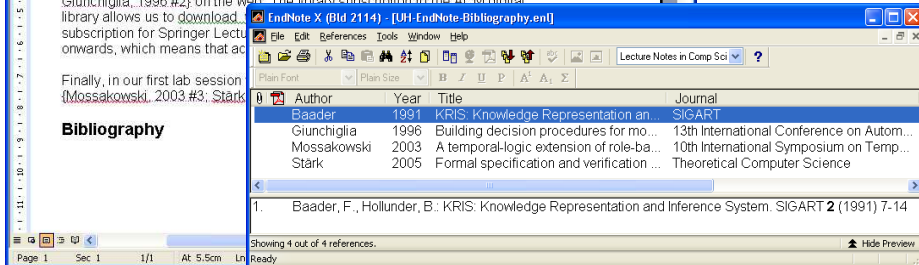
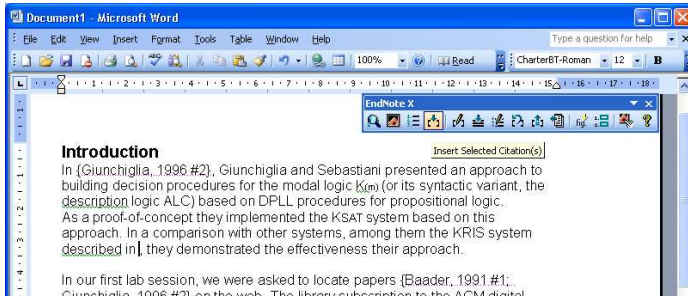
Author	Year	Title	Journal
Baader	1991	KRIS: Knowledge Representation an...	SIGART
Giunchiglia	1996	Building decision procedures for mo...	13th International Conference on Autom...
Mossakowski	2003	A temporal-logic extension of role-ba...	10th International Symposium on Temp...
Stärk	2005	Formal specification and verification ...	Theoretical Computer Science

1. Mossakowski, T., Drouineaud, M., Sohr, K.: A temporal-logic extension of role-based access control covering dynamic separation of duties. 10th International Symposium on Temporal Representation and Reasoning / 4th International Conference on Temporal Logic (TIME-ICTL 2003). IEEE Computer Society (2003) 83-90

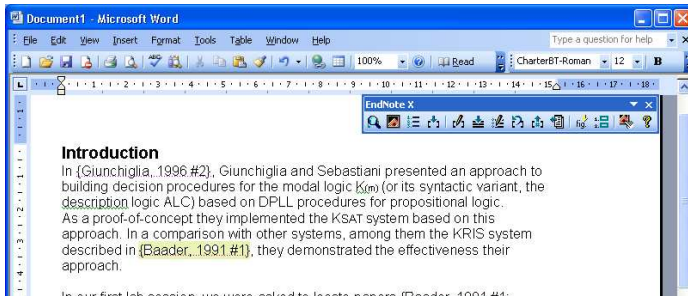
Showing 4 out of 4 references. Hide Preview

Ready

EndNote: Interacting with Word (1)



EndNote: Interacting with Word (2)

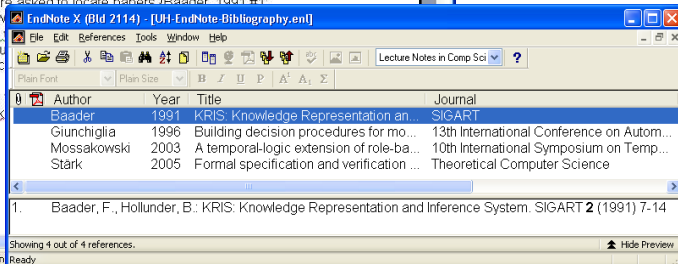


In our first lab session, we were asked to locate papers {Baader, 1991 #1;

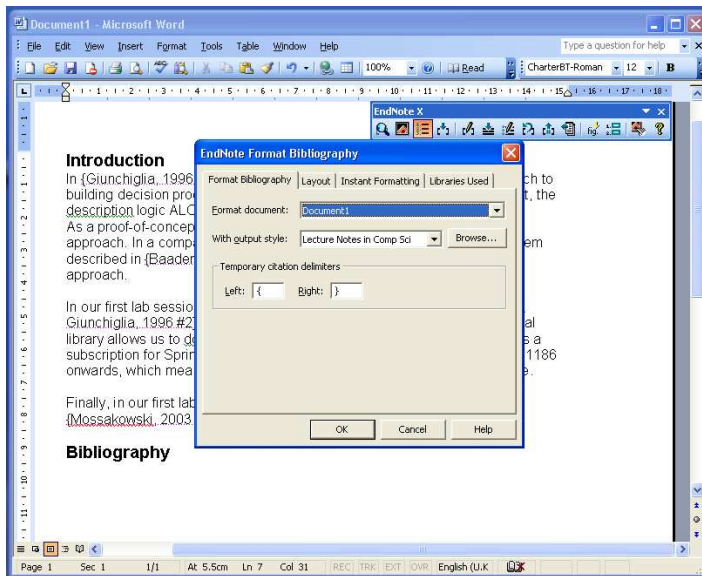
Giunchiglia, 1996 #2} on the w library allows us to download... subscription for Springer Lectu onwards, which means that ac

Finally, in our first lab session {Mossakowski, 2003 #3; Stark

Bibliography



EndNote: Creating a bibliography (1)



EndNote: Creating a bibliography (2)

Introduction

In [2], Giunchiglia and Sebastiani presented an approach to building decision procedures for the modal logic K_m (or its syntactic variant, the description logic ALC) based on DPLL procedures for propositional logic. As a proof-of-concept they implemented the KSAT system based on this approach. In a comparison with other systems, among them the KRIS system described in [1], they demonstrated the effectiveness their approach.

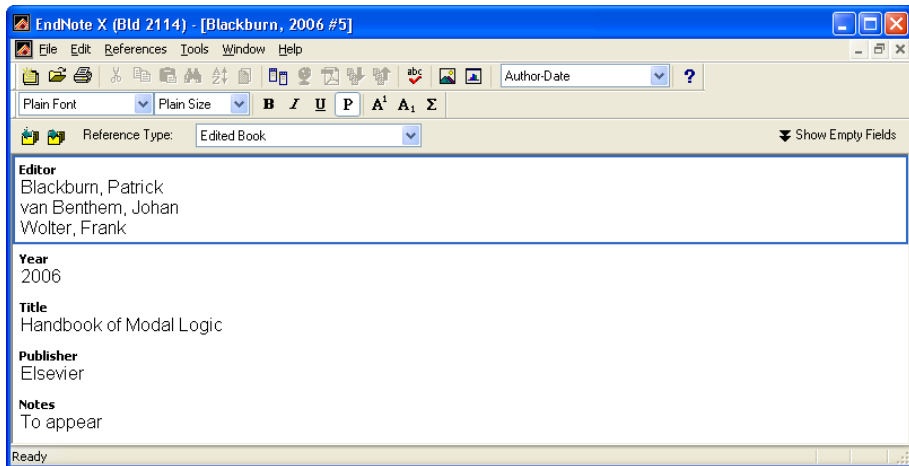
In our first lab session, we were asked to locate papers [1, 2] on the web. The library subscription to the ACM digital library allows us to download without charge. However, the library only has a subscription for Springer Lecture Notes in Computer Science from volume 1186 onwards, which means that access to articles in LNAI 1104 incurs a charge.

Finally, in our first lab session we were also asked to locate papers [3, 4].

Bibliography

1. Baader, F., Hollunder, B.: KRIS: Knowledge Representation and Inference System. SIGART **2** (1991) 7-14
2. Giunchiglia, F., Sebastiani, R.: Building decision procedures for modal logics from propositional decision procedures - the case study of modal K. 13th International Conference on Automated Deduction (CADE-13), Vol 1104. Springer (1996) 583-597
3. Mossakowski, T., Drouineaud, M., Sohr, K.: A temporal-logic extension of role-based access control covering dynamic separation of duties. 10th International Symposium on Temporal Representation and Reasoning / 4th International Conference on Temporal Logic (TIME-ICTL 2003). IEEE Computer Society (2003) 83-90
4. Stark, R.F.: Formal specification and verification of the C# thread model. Theoretical Computer Science **343** (2005) 482-508

EndNote: Entering more references



EndNote: Interacting with Word (3)

Document1 - Microsoft Word

File Edit View Insert Format Tools Table Window Help

Type a question for help

100%

CharterBT-Roman 12

Introduction

In [2], Giunchiglia and Sebastiani presented an approach to building decision procedures for the modal logic K_m (or its syntactic variant, the description logic ALC) based on DPLL procedures for propositional logic. As a proof-of-concept they implemented the KSAT system based on this approach. In a comparison with other systems, among them the KRIS system described in [1], they demonstrated the effectiveness their approach.

In our first lab session, we used the ACM digital library allows us to search for Springer Lecture Notes in Computer Science articles to articles in LNAI 1104 including

Finally, in our first lab session we presented the following results:

Bibliography

1. Baader, F., Hollunder, T., & Rindler-Schjerve, F. (1996). Building decision procedures - the case study of LNAI 1104. Springer (1996) 583-597
2. Giunchiglia, F., & Sebastiani, R. (1996). Building decision procedures for modal logic. 13th International Conference on Automated Reasoning in Artificial Intelligence (1996) 583-597
3. Mossakowski, T., & Drovse, S. (2003). A temporal-logic extension of role-based temporal logic. 10th International Symposium on Temporal Representation and Reasoning (2003) 482-508
4. Stark, R.F. Formal specification and verification of temporal logic. Theoretical Computer Science (2005) 482-508

EndNote X (Bld 2114) - [UH-EndNote-Bibliography.enl]

File Edit References Tools Window Help

Lecture Notes in Comp Sci

Author	Year	Title	Journal
Baader	1991	KRIS: Knowledge Representation and Reasoning	SIGART
Blackburn	2006	Handbook of Modal Logic	
Giunchiglia	1996	Building decision procedures for modal logic	13th International Conference on Automated Reasoning in Artificial Intelligence
Mossakowski	2003	A temporal-logic extension of role-based temporal logic	10th International Symposium on Temporal Representation and Reasoning
Stark	2005	Formal specification and verification of temporal logic	Theoretical Computer Science

Showing 5 out of 5 references.

EndNote: Interacting with Word (4)

The screenshot shows a Microsoft Word 2003 window titled 'Document1 - Microsoft Word'. The EndNote X toolbar is visible, containing icons for inserting references, tables, and other EndNote functions. The document content is as follows:

Introduction

In [2], Giunchiglia and Sebastiani presented an approach to building decision procedures for the modal logic K_m (or its syntactic variant, the description logic ALC) based on DPLL procedures for propositional logic. As a proof-of-concept they implemented the KSAT system based on this approach. In a comparison with other systems, among them the KRIS system described in [1], they demonstrated the effectiveness their approach.

In our first lab session, we were asked to locate papers [1, 2] on the web. The library subscription to the ACM digital library allows us to download without charge. However, the library only has a subscription for Springer Lecture Notes in Computer Science from volume 1186 onwards, which means that access to articles in LNAI 1104 incurs a charge.

Finally, in our first lab session we were also asked to locate papers [3, 4] as well as the forthcoming {Blackburn, 2006 #5}.

Bibliography

1. Baader, F., Hollunder, B.: KRIS: Knowledge Representation and Inference System. SIGART **2** (1991) 7-14
2. Giunchiglia, F., Sebastiani, R.: Building decision procedures for modal logics from propositional decision procedures - the case study of modal K. 13th International Conference on Automated Deduction (CADE-13), Vol. 1104. Springer (1996) 583-597
3. Mossakowski, T., Drouineaud, M., Sohr, K.: A temporal-logic extension of role-based access control covering dynamic separation of duties. 10th International Symposium on Temporal Representation and Reasoning / 4th International Conference on Temporal Logic (TIME-ICTL 2003). IEEE Computer Society (2003) 83-90
4. Stark, R.F.: Formal specification and verification of the C# thread model. Theoretical Computer Science **343** (2005) 482-508

The status bar at the bottom indicates 'Page 1', 'Sec 1', '1/1', 'At: 8.9cm', 'Ln 14', 'Col 34', and the language is set to 'English (U.K.)'.

EndNote: Updating a bibliography (1)

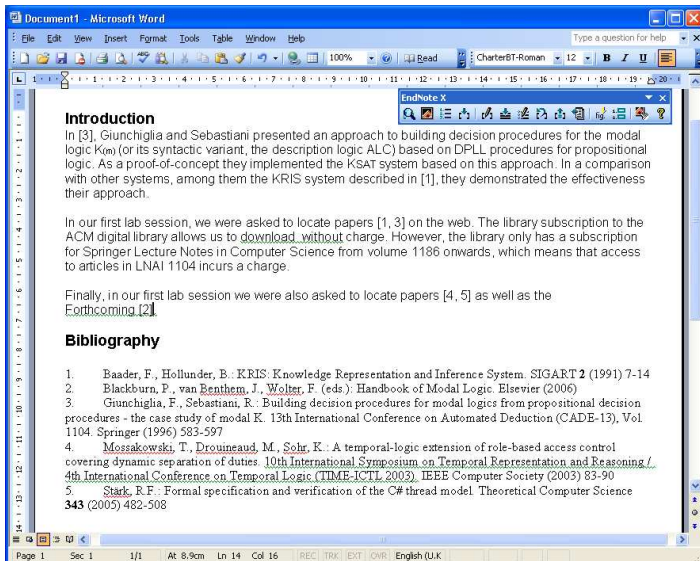
The screenshot shows a Microsoft Word document titled "Document1 - Microsoft Word". The document contains an "Introduction" section and a "Bibliography" section. The "EndNote Format Bibliography" dialog box is open, displaying the "Format Bibliography" tab. The dialog box has the following settings:

- Format document: Document1
- With output style: Lecture Notes in Comp Sci
- Temporary citation delimiters: Left: {, Right: }

The "Bibliography" section of the document lists the following references:

1. Baader, F., Hollunder, M., & Giunchiglia, F. (1996). Procedures for the modal logic K_m (or its syntactic logic). As a proof-of-concept with other systems, among their approach.
2. Giunchiglia, F., Sebastiani, R. (1996). Procedures for the modal logic K_m (or its syntactic logic). As a proof-of-concept with other systems, among their approach.
3. Mossakowski, T., Drouineau, M., Sohr, K.: A temporal-logic extension of role-based access control covering dynamic separation of duties. 10th International Symposium on Temporal Representation and Reasoning / 4th International Conference on Temporal Logic (TIME-ICTL 2003). IEEE Computer Society (2003) 83-90
4. Stark, R.F.: Formal specification and verification of the C# thread model. Theoretical Computer Science 343 (2005) 482-508

EndNote: Updating a bibliography (2)

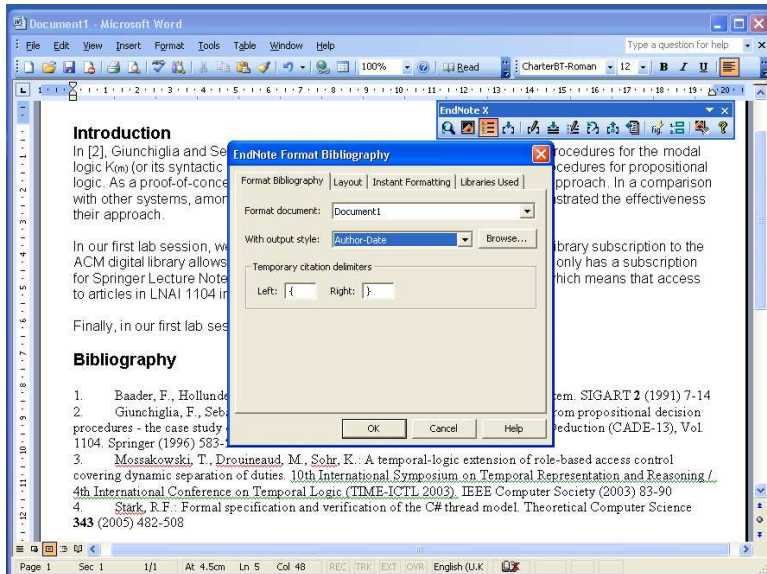


The screenshot shows a Microsoft Word document titled "Document1 - Microsoft Word". The document contains an "Introduction" section and a "Bibliography" section. The "Bibliography" section lists five references, each preceded by a number. The references are:

1. Baader, F., Hollunder, B.: KRIS: Knowledge Representation and Inference System. SIGART **2** (1991) 7-14
2. Blackburn, P., van Benthem, J., Wolter, F. (eds.): Handbook of Modal Logic. Elsevier (2006)
3. Giunchiglia, F., Sebastiani, R.: Building decision procedures for modal logics from propositional decision procedures - the case study of modal K. 13th International Conference on Automated Deduction (CADE-13), Vol. 1104. Springer (1996) 583-597
4. Mossakowski, T., Drouineaud, M., Sohr, K.: A temporal-logic extension of role-based access control covering dynamic separation of duties. 10th International Symposium on Temporal Representation and Reasoning / 4th International Conference on Temporal Logic (TIME-ICTL 2003). IEEE Computer Society (2003) 83-90
5. Stark, R.F.: Formal specification and verification of the C# thread model. Theoretical Computer Science **343** (2005) 482-508

The document also includes an "Introduction" section with three paragraphs. The first paragraph discusses the approach of Giunchiglia and Sebastiani. The second paragraph discusses the library subscription to the ACM digital library. The third paragraph discusses the location of papers [4, 5] and the forthcoming [2].

EndNote: Re-formatting a bibliography (1)



EndNote: Re-formatting a bibliography (2)

Document1 - Microsoft Word

File Edit View Insert Format Tools Table Window Help

Type a question for help

100% Read

CharterBT-Roman 12 B I U

EndNote X

Introduction

In (Giunchiglia and Sebastiani 1996), Giunchiglia and Sebastiani presented an approach to building decision procedures for the modal logic K_m (or its syntactic variant, the description logic ALC) based on DPLL procedures for propositional logic. As a proof-of-concept they implemented the KSAT system based on this approach. In a comparison with other systems, among them the KRIS system described in (Baader and Hollunder 1991), they demonstrated the effectiveness their approach.

In our first lab session, we were asked to locate papers (Baader and Hollunder 1991; Giunchiglia and Sebastiani 1996) on the web. The library subscription to the ACM digital library allows us to download without charge. However, the library only has a subscription for Springer Lecture Notes in Computer Science from volume 1186 onwards, which means that access to articles in LNAI 1104 incurs a charge.

Finally, in our first lab session we were also asked to locate papers (Mossakowski, Drouineaud et al. 2003; Stark 2005).

Bibliography

Baader, F. and B. Hollunder (1991). "KRIS: Knowledge Representation and Inference System." *SIGART* 2(3): 7-14.

Giunchiglia, F. and R. Sebastiani (1996). *Building decision procedures for modal logics from propositional decision procedures - the case study of modal K*. 13th International Conference on Automated Deduction (CADE-13). Springer.

Mossakowski, T. M. Drouineaud, et al. (2003). *A temporal-logic extension of role-based access control covering dynamic separation of duties*. 10th International Symposium on Temporal Representation and Reasoning / 4th International Conference on Temporal Logic (TIME-ICTL 2003), IEEE Computer Society.

Stark, R. F. (2005). "Formal specification and verification of the C# thread model." *Theoretical Computer Science* 343(3): 482-508.

Page 1 Sec 1 1/1 At 4.5cm Ln 5 Col 88 REC ITRK EXT OVR English (U.K.)

Conclusion

- Tools like EndNote help you to maintain a large set of bibliographic references
- They ease the burden of referencing and generating lists of references according to a specific style
- If no specific style is requested, then a providing all the necessary information about each of your sources in a **consistent way** is the most important aspect of a bibliography
- Beware that the way you formulate sentences which include references depends on the referencing style; changing that style later on is time-consuming and error-prone

Research Methods in Computer Science

Lecture 13: Acknowledging Your Sources

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Previously ...

26 Bibliography styles

- Ordinal number
- Author-date
- Abbreviation

27 Using EndNote for Citations and Bibliographies

- Introduction
- References
- Word
- Conclusion

Today ...

28 Citing

29 Quoting

30 Support

Citing: Rules of thumb (Zobel 2004)

- If you discuss a paper in detail or note some particular contribution it makes, it **must** be cited
- Claims, statements of fact, discussions of previous work **should** be supported by references, if not supported by your current work
But: Do not cite to support common knowledge;
do not end every sentence with a reference
- References to your own previous work is **allowed** if it is relevant to your current work
But: Gratuitous self-reference is counterproductive
- Attribute work correctly, in particular, when relying on secondary sources
Bad: According to Dawson (1981), stable graphs have been shown to be closed
Good: According to Kelly (1959; as quoted by Dawson, 1981), stable graphs are closed

Citing (1)

Original text [19]:	Student's text:
KNOWITALL is an autonomous system that extracts facts, concepts, and relationships from the web. KNOWITALL is seeded with an extensible ontology and a small number of generic rule templates from which it creates text extraction rules for each class and relation in its ontology. The system relies on a domain- and language-independent architecture to populate the ontology with specific facts and relations.	An example of the described system is KNOWITALL [19]. It is an autonomous system that extracts facts, concepts, and relationships from the web. KNOWITALL [19] is seeded with an extensible ontology and a small number of generic rule templates from which it creates text extraction rules for each class and relation in its ontology. The system relies on a domain- and language-independent architecture to populate the ontology with specific facts and relations.

References are **not** meant to indicate copying! This is **wrong!**

Citing (1)

Original text [19]:	Improved text:
KNOWITALL is an autonomous system that extracts facts, concepts, and relationships from the web. KNOWITALL is seeded with an extensible ontology and a small number of generic rule templates from which it creates text extraction rules for each class and relation in its ontology. The system relies on a domain- and language-independent architecture to populate the ontology with specific facts and relations.	An example of the described system is KNOWITALL [19]. Given an initial ontology and a small number of rule templates which do not depend on the class and relationships in the ontology, KNOWITALL generates text extraction rules for each class and relationship in the ontology. These text extraction rules are then applied to texts found on the web. Rule applications populate the ontology with instances of the concepts and relationships in the ontology.

Citing (2)

- A reference in ordinal-number style never starts a sentence

Wrong: [9] Disaster rescue is a serious social issue.

Correct: Disaster rescue is a serious social issue [9].

- In ordinal-number style a list of references is a comma-separated list of numbers enclosed in one pair of square brackets

Wrong: The humanoid soccer robots are fully autonomous [5][9].

Correct: The humanoid soccer robots are fully autonomous [5,9].

- A reference never occurs in a section heading

Wrong: Section 5. The History of RoboCup [9]

Wrong: Section 5. The History of RoboCup (Henry 2006)

Citing (2)

- A reference never comes after a full stop

Wrong: 2-on-2 teams of autonomous mobile robots play games in a rectangular field color-coded in shades of grey. [9]

Correct: 2-on-2 teams of autonomous mobile robots play games in a rectangular field colour-coded in shades of gray [9].

- Beware of the differences between ordinal-number style and author-date style

Wrong: [11,12] stresses the importance of algorithmic support for formal methods.

Correct: Wolper (1996a, 1996b) stresses the importance of algorithmic support for formal methods.

Correct: Wolper [11,12] stresses the importance of algorithmic support for formal methods.

Citing (3)

Examples of correct use of author-date style:

While Wolper (1996a) does not argue that compositionality in proof systems for concurrency is undesirable, he claims that achieving it without algorithmic support is mostly useless.

Recent work (Wolper 1996a, 1996b) stresses the importance of algorithmic support for formal methods.

Wolper (1996a, 1996b) stresses the importance of algorithmic support for formal methods.

The completion procedure may fail in general, but has been extended to a refutationally complete theorem prover (cf. Lankford 1975, Hsiang and Rusinowitch 1987).

Completion procedures for conditional equations have been described by Kounalis and Rusinowitch (1988), and by Ganzinger (1987a, 1987b).

Quoting

Example taken from a student's text:

Such dangers are catered for by ensuring the closure of the function set. Koza [1992] states that:

The closure property requires that each of the functions in the function set be able to accept, as its arguments, any value and data type that may possibly be assumed by any terminal set. That is, each function in the function set should be well defined and closed for any combination of arguments that it may encounter.

Without closure, many individuals could have their fitness drastically lowered as a result of minor syntactic errors.

Direct quotation from Koza [1992]; clearly indicated as such; restricted to (less than) one paragraph; source stated.

Quoting

Examples taken from a student's text:

Bickle [1996] states that “[t]he superior method to obtain compact and accurate solutions is the method of adaptive parsimony pressure [...]”.

Quotation clearly indicated by quotation marks; alterations indicated in square brackets; source stated.

Day [2005] reports that “GP shows great promise in creating robust classifiers for [Automatic Speaker Verification] purposes” where programs attempt to recognise the voice of a known individual.

Quotation clearly indicated by quotation marks; alterations indicated in square brackets; source stated.

Quoting

Examples taken from a student's text:

More recently, in 1999, Tim Berners-Lee [3], father of the World Wide Web (WWW) speaking of the WWW stated that he saw it as “an information space through which people can communicate; but communicate in a special way: communicate by sharing their knowledge in a pool. The idea was not that it should be a big browsing medium. The idea was that everybody would be putting their ideas in as well as taking them out.”

A Wiki is in Ward Cunningham's [43] original description: “The simplest online database that could possibly work.”

Direct quotation indicated by quotation marks and indentation; source stated.

Quoting

- Avoid excessive quotation.
- Quotation is only appropriate
 - where you want to comment on the statements made by someone else
 - where the quote is of some historical significance
- In all other cases, use your own words

Evidence and Support

Example taken from a student's text:

Intelligent agents, autonomous or semi-autonomous systems that take decisions and perform tasks in complex, dynamically changing environments, revolutionized the field of AI.

- This is stating an **opinion** not a generally known and accepted **fact**
- As such it needs **support** which it currently lacks
- **Support** could be provided by some **statistical evidence** or by a **reference**
- (Made-up) example of **statistical evidence**:

The concept of intelligent agents was first introduced in 1983. By 2003, more than half of all papers published in the main forums of AI, referred to the concept or made use of intelligent agents, and it has spawned a world wide industry worth 5 billion US\$ [2].

where [2] is a reference to the source of these statistics.

Evidence and Support

Example taken from a student's text:

To deal with information in the web environment what is needed is a logic that supports modes of reasoning which are approximate rather than exact.

- Again, this is stating an **opinion** not a generally known and accepted **fact**
- **Support** could be provided by an **argument** or by a **reference**
- A **reference** could point to a scientific paper where this opinion is stated and argued for
- An **argument** could be an **example** illustrating the advantage of approximate over exact reasoning
If that example is taken from a source, then again that source needs to be referenced

Evidence and Support

Example taken from a student's text:

To deal with information in the web environment what is needed is a logic that supports modes of reasoning which are approximate rather than exact.

Better formulation:

It has been argued by Oberschlau [1] that to deal with information in the web environment what is needed is a logic that supports modes of reasoning which are approximate rather than exact.

or

According to Oberschlau [1], to deal with information in the web environment what is needed is a logic that supports modes of reasoning which are approximate rather than exact.

Evidence and Support

Example taken from a student's text:

Therefore, once our system is enhanced with our common knowledge about things we know, [it] could be seen as an intelligent entity. A brilliant example is the Cyc knowledge base.

- The phrase 'brilliant example' is ambiguous:
'Cyc a system incorporating common knowledge and it is a good example of such a system' versus
'Cyc is a brilliant system incorporating common knowledge'
- Both readings require support, in particular, the second version

Evidence and Support

- As Lenat (1995) has noted in an earlier paper, Cyc is a brilliant system.
- Lenat (1995) demonstrates that Cyc is a brilliant system.
- Cyc is a brilliant system (Lenat 1995).

In the sentences above, the author **agrees** with Lenat (1995)

- Lenat (1995) alleges that Cyc is a brilliant system.
- Lenat (1995) claims that Cyc is a brilliant system.

In the sentences above, the author **disagrees** with Lenat (1995)

- Lenat (1995) states that Cyc is a brilliant system.

In the sentence above, the author is **neutral** with regard to the truth of the statement 'Cyc is a brilliant system'

Evidence and Support

Example taken from a student's text:

The most popular ways to compress data are the Huffman coding and Shannon-Fano coding.

- It is unclear on what basis compression methods are judged to be 'popular'
 - number of compressed files
 - number of users of compression software
 - number of developers of compression software
- In each case, **statistical** evidence seems to be required, e.g.

In 2004, 60% of all compressed files were compressed using the Huffman coding or Shannon-Fano coding [3].

where [3] is reference to the source of these statistics.

Research Methods in Computer Science

Lecture 14: Presentations and Presentation Skills

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Presentations: Recall

- People remember
 - 20% of what they hear
 - 30% of what they see/read
 - 50% of what they see and hear
 - 70% of what they say and write
 - 90% of what they do
- Regarding information presented during a (one hour) lecture, students retain
 - 70% of the first 10 minutes
 - 20% of the last 10 minutes
- Are there **techniques** that can help us improve the recall of the audience or at least focus their recall on the important aspect of a presentation?

Today's questions

- 1 What different types of presentations can you think of?
- 2 What is the typical structure of a presentation?
- 3 What steps do you go through when preparing a presentation?

(15 minutes group discussion)

Types of presentations

Presentations typically serve one or more of the following purposes:

Purpose: Information delivery, Information gathering, Instruction, or Persuasion

In addition, we can classify presentations along the following 'scales':

Medium: Verbal, Verbal with Visual Aids, or Written

Presence: In person — Transmitted — Recorded

Interaction: Monolog — Dialogue

Time: Short — Long

Audience: Small — Large

Setting: Informal — Formal

Preparation: Ad hoc — Scripted

Structure of presentations

- Introduction
 - ~> motivation, contextualisation, overview
- Main body
 - ~> main findings, elaboration
- Conclusion
 - ~> comment on importance of findings, future work, summary

Preparing presentations

- 1 Determination of the objectives of the presentation
- 2 Analysis of the audience
- 3 Planning
- 4 Organisation of the material for effective results
- 5 Preparation of visual aids / handouts
- 6 Delivery practice

Research Methods in Computer Science

Lecture 15: Presentations and Presentation Skills (2)

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Previously ...

31 Presentations

- Types
- Structure
- Preparation

Today's questions

- 1 What is the purpose of visual aids?
- 2 What types of visual aids do you know?
- 3 Can you give a 'style guide' for slides?

Focus on the overall structure of a slide
(Other aspects will be covered in the following lectures)

(10 minutes group discussion)

Visual aids: Purpose

- Give structure to a presentation
- Provide a point of reference for the speaker and the audience
- Help an audience to remember
- Focus the attention of both audience and speaker
- Reinforce what is said

Visual aids: Types

- Prepared in advance, immutable at time of presentation
 - Video
 - Slide projector
- Created or reproduced during presentation, mutable at time of presentation
 - Flip chart
 - Chalk/White board
- Dual use
 - Overhead projector (OHP)
 - LCD projector (beamer) plus PC
 - Interactive white board plus PC

Slides: Structure

- Decide on a **structure** / **theme** for your slide in advance, then stick to it
- Consider the following questions:
 - Does the audience know me (and my affiliation)?
 - How important is it that the audience remembers the title of my presentation?
 - How many navigational hints are required?
 - How many graphics do I need to include? Can they be placed consistently?

The answers to these questions influence how you should structure your slides

Structure: Example (1)

There Is No Largest Prime Number
 Euklid

Results
 Proof of a particular Theorem

There Is No Largest Prime Number

With an introduction to a new proof technique

Euklid of Alexandria
 Department of Mathematics
 University of Alexandria

27th International Symposium on Prime Numbers, –280

1

Results

- Proof of the Main Theorem

There Is No Largest Prime Number
 The proof uses *reductio ad absurdum*.

Results
 Proof of the Main Theorem

Theorem

There is no largest prime number.

Proof.

- 1 Suppose p were the largest prime number.
- 2 Let q be the product of the first p numbers.
- 3 Then $q + 1$ is not divisible by any of them.
- 4 Thus $q + 1$ is also prime and greater than p . □

Structure: Example (2)

There Is No Largest Prime Number

With an introduction to a new proof technique

Euklid of Alexandria

Department of Mathematics
University of Alexandria

27th International Symposium on Prime Numbers,
—280

Results

Proof of the Main Theorem

There Is No
Largest Prime
Number
Euklid

Results
Proof of the Main Theorem

There Is No Largest Prime Number

The proof uses *reductio ad absurdum*.

Theorem

There is no largest prime number.

Proof.

1. Suppose p were the largest prime number.
2. Let q be the product of the first p numbers.
3. Then $q + 1$ is not divisible by any of them.
4. Thus $q + 1$ is also prime and greater than p . □

There Is No
Largest Prime
Number
Euklid

Results
Proof of the Main Theorem

Structure: Example (3)

Results
○

There Is No Largest Prime Number

With an introduction to a new proof technique

Euklid of Alexandria
Department of Mathematics
University of Alexandria

27th International Symposium on Prime Numbers, –280

Results
Proof of the Main Theorem

Euklid
There Is No Largest Prime Number

Univ. Alexandria

Results
●
Proof of the Main Theorem

There Is No Largest Prime Number

The proof uses *reductio ad absurdum*.

Theorem

There is no largest prime number.

Proof.

1. Suppose p were the largest prime number.
2. Let q be the product of the first p numbers.
3. Then $q + 1$ is not divisible by any of them.
4. Thus $q + 1$ is also prime and greater than p . □

Euklid
There Is No Largest Prime Number

Univ. Alexandria

Structure: Example (4)

There Is No Largest Prime Number

With an introduction to a new proof technique

Euklid of Alexandria

Department of Mathematics
University of Alexandria

27th International Symposium on Prime Numbers, –280

1 Results

- Proof of the Main Theorem

Euklid (Univ. Alexandria)

There Is No Largest Prime Number

ISPN '80 1 / 2

There Is No Largest Prime Number

The proof uses *reductio ad absurdum*.

Theorem

There is no largest prime number.

Proof.

- 1 Suppose p were the largest prime number.
- 2 Let q be the product of the first p numbers.
- 3 Then $q + 1$ is not divisible by any of them.
- 4 Thus $q + 1$ is also prime and greater than p . □

Euklid (Univ. Alexandria)

There Is No Largest Prime Number

ISPN '80 2 / 2

Research Methods in Computer Science

Lecture 16: Presentations and Presentation Skills (3)

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University of Liverpool

Previously ...

31 Presentations

- Types
- Structure
- Preparation

Today's questions

1 Can you give a 'style guide' for slides?

Consider

- Title
- Textual content
- Fonts
- Colours
- Graphics and animations

(10 minutes group discussion)

Slides: Titles

- Put a title on each slide
- Titles should be short but **descriptive**
- Ideally, titles on consecutive slides should **tell a story** all by themselves
- **Capitalise** words **consistently**
 - Either always capitalise all words in the title (except for words like 'a' and 'the'), or
 - always only capitalise the first word in the title/subtitle
- The title of the whole presentation should be capitalised
You might want to include it on every slide

Slides: Textual content (1)

- Keep it simple
- A typical slide should contain 20 to 40 words, maximum 80
- Do **not** try to fill all the space
- Prefer enumerated or itemised **lists** over plain text
- Use **at most two levels** of 'subitemizing'
- Keep the **number of items** in a list **low**
- Highlight important things

Slides: Textual content (2)

- Use **short sentences**
- Prefer **phrases** over complete sentences
- Break lines where there is a logical pause
- **Do not hyphenate** words
- **Punctuate consistently**
 - No punctuation after phrases
 - Complete punctuation in and after complete sentences
- **Avoid decreasing font size** to make more text fit on a slide

Slides: Fonts

- Aim for your text to be **legible** even under difficult conditions
- Use as few fonts as possible
- Use a **sans-serif font** unless you use a high-resolution LCD projector
- Use **monospaced** and **script fonts** only for specific purposes
- **Avoid italics** to express emphasis, **use colour instead**

Slides: Colours

- Use colours sparsely
- Avoid bright text on dark background
- Maximise contrast
 - Normal text should be black on (nearly) white background
 - Avoid bright, light colours on white background
- Be aware of what we associate with different colours
- Test your presentation on the intended equipment if possible

Slides: Colour associations

Red	Danger, aggression, passion, stimulating
Purple	Royalty, religion, calming
Green	Soothing, trustworthiness, nature
Blue	Restful, peaceful, relaxing
Yellow	Well-being
Brown	Nature, practicality, boring, close minded

Decide

- what you want to highlight,
e.g. keywords, main results, examples, current focus
- which colour you want to use for each of these categories

Then apply this **colour scheme** consistently

Slide structure and content: Graphics and animations

- **Graphics** often convey concepts or ideas more effectively than text
- Use graphics as often as possible
- **Graphics** should only contain as much detail as necessary
- **Graphics** always require **explanation**
- Use **animations** to explain the **dynamics of systems, algorithms, ...**
- Do **not** use animations to simply attract attention
- Do **not** use distracting **special effects** like fancy slide transitions

Research Methods in Computer Science

Lecture 17: Presentations and Presentation Skills (4)

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Department of Computer Science
University of Liverpool

Previously ...

34 Slides

- Titles
- Content
- Fonts
- Colours
- Graphics

Today's questions

- 1 How should you behave during a presentation?
- 2 What kind of behaviour should you avoid during a presentation?

Consider

- Stance
- Hands
- Eye contact
- Voice

(12 minutes group discussion)

Gesture and Body Language: Stance and Movement

- Be aware where you stand (centre stage vs side stage)
- Do not obscure the screen
- Stand tall, keep your head up most of the time
- Move from stillness to stillness, walk slowly

Gesture and Body Language: Hands

- Use hand gestures to emphasise points
- Use open palm gestures, full arm gestures
- **Avoid** aggressive gestures
- **Avoid** hands in pockets, hands behind your back, hands clasped in front of your body

Gesture and Body Language: Eye Contact

- Maintain eye contact
 - lighthouse beam
 - treat everyone equal
 - do not look out of the window or on your watch
 - do not focus too long on a single individual
- Keep an eye on the audience's body language
 - does a point need further clarification?
 - can you proceed more quickly than anticipated?

Gesture and Body Language: Voice

- Be aware of the **acoustics of the room**
- **Speak clearly** (do not shout or whisper)
- **Pause** shortly at key points (adds emphasis)
- **Emphasise the right words**, control your breathing
- **Facial gestures** and tone of voice should match your message
- Do **not** rush, or talk deliberately slowly, but vary speed
- Do **not** talk to the screen
- Do **not** turn your back to the audience and talk at the same time
- Do **not** read from a script (cue cards are ok)

Seven Principles of Public Speaking

(Isa N. Engleberg: The Principles of Public Presentation.
Harper Collins, New York, 1994)

Purpose: Why are you speaking?

What do you want audience members to know, think, believe, or do as a result of your presentation?

People: Who is your audience?

How do the characteristics, skills, opinions, and behaviours of your audience affect your purpose?

Place: How can you plan and adapt to the logistics of this place?
How can you use visual aids to help you achieve your purpose?

Seven Principles of Public Speaking

(Isa N. Engleberg: The Principles of Public Presentation.
Harper Collins, New York, 1994)

Preparation: Where and how can you find good ideas and information for your speech?

How much and what kind of supporting materials do you need?

Planning: Is there a natural order to the ideas and information you will use?

What are the most effective ways to organise your speech in order to adapt it to the purpose, people, place, etc.?

Seven Principles of Public Speaking

(Isa N. Engleberg: The Principles of Public Presentation.
Harper Collins, New York, 1994)

Personality: How do you become associated with your message
in a positive way?

What can you do to demonstrate your competence,
charisma, and character to the audience?

Performance: What form of delivery is best suited to the purpose of
your speech?

What delivery techniques will make your presentation
more effective?

How should you practice?

Research Methods in Computer Science

Lecture 18: Choosing or proposing a project (1)

Ullrich Hustadt

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Previously ...

35 Gesture and Body Language

- Stance
- Hands
- Eye Contact
- Voice

36 Principles of Public Speaking

Topics

- 37 Choosing a project
 - Sources of information
 - Criteria
 - Suitability criteria

Today's questions

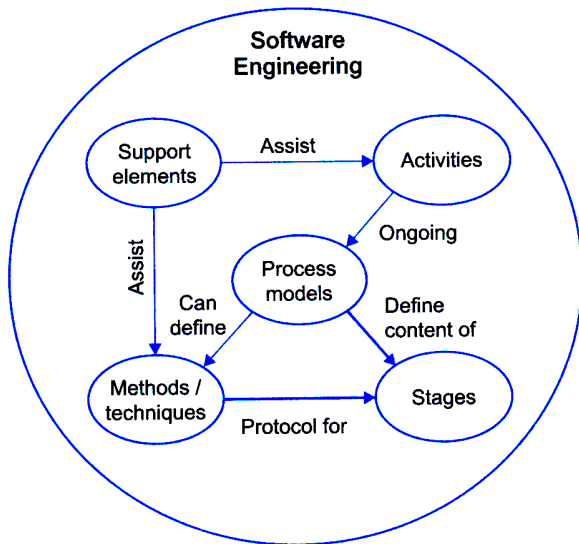
- 1 What **sources of information** could be used to devise a research-oriented project?
- 2 Given a collection of proposals for research-oriented projects, what **criteria** could you use to select the most suitable one?

(10 minutes group discussion)

Sources of information

- Proposals by academic staff or departments
- Past projects
- Brainstorming
- Your own goals and learning objectives
- Reading about / working in the subject area
- Systematic analysis of the subject area
 - **Research Territory Maps**
Show how topics related to each other
 - **Relevance Trees**
Break down a particular subject or research question into lower and lower levels of detail
 - **Spider Diagrams**
Combines features of Research Territory Maps with those of Relevance Trees

Research Territory Maps: Example



Relevance Trees: Example

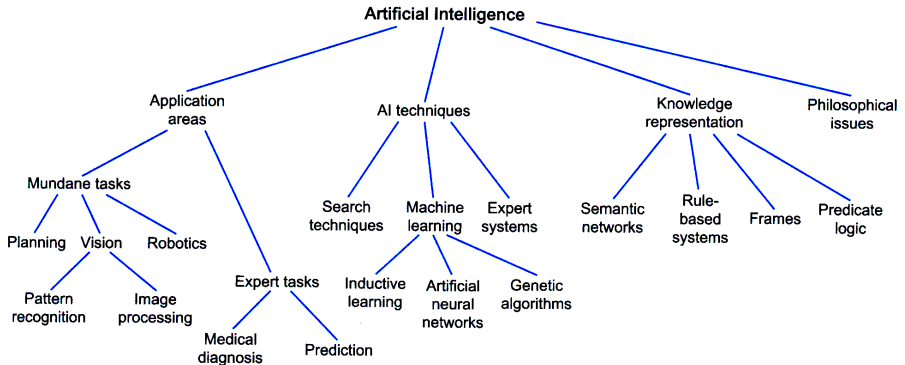


Figure 3.2 An example relevance tree for the field of *artificial intelligence*

Spider Diagrams: Example

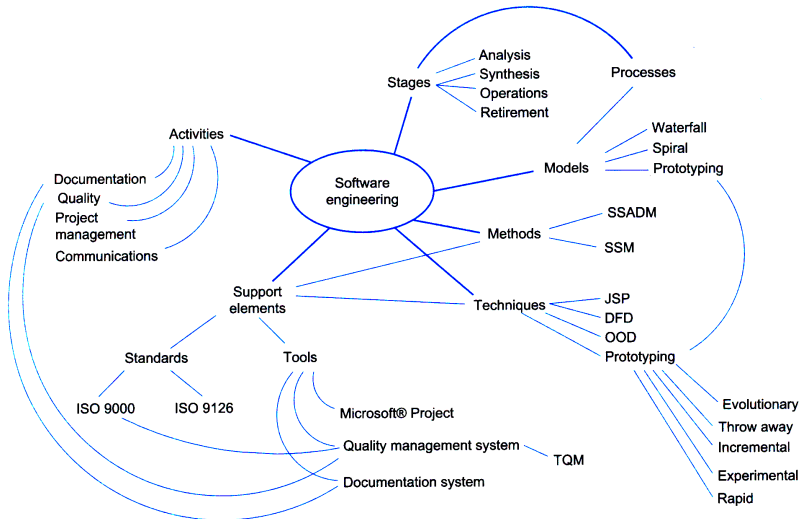


Figure 3.3 An example spider diagram for the field of *software engineering*

Choosing a project

- The project needs to be within your **capabilities**
- The project needs to have **sufficient scope**
- The project needs to **interest** you
- The project needs to have a **serious purpose**
- The project needs to have a **clear outcome**
- The project needs to be **related to your degree programme**
- The **resources** required for the project are available or can be obtained

Suitability tests for projects

- 'So what?' test

Is the topic meaningful?

Will it be of value for anyone?

What contribution will it make?

- Justification

Can you explain your project and justify it in simple terms?

- Estimating your understanding

Can you put a figure on what you know about your chosen subject?

- Contacts

Are the contacts you require for your project (including your supervisor) available, accessible, and willing to help?

- Project proposal

Can you write a substantive proposal for your project?

Research Methods in Computer Science

Lecture 19: Choosing or proposing a project (2)

Ullrich Hustadt

Department of Computer Science
University of Liverpool

Previously ...

- 37 Choosing a project
 - Sources of information
 - Criteria
 - Suitability criteria

Today's questions

Take the project proposal you were given last week as a reference to answer the following two questions:

- ① What is the (implicit) content of a project proposal?
What kind of questions does it need to address?
- ② What is the explicit structure of a project proposal?
What sections/parts are there? What is their purpose?

(10 minutes group discussion)

Preparing a project proposal: Implicit Content

- Introduction to the subject area
 - Sets the context for the project
 - Should motivate the relevance of the subject area
- Overview of current research in the area
 - Demonstrates current activities in the subject area
 - Shows your understanding of current research
- Identify a gap
 - Identify a need for further investigation or re-interpretation
- Identify how your work fills the gap
 - Explain how your project fills the gap
- Identify risks and solutions
 - Highlight the benefits that can be derived from your project
 - Account for the risks to your project

Preparing a project proposal: Explicit Structure (1)

- Title

Clear, Concise, Preferably no acronyms

- Aims and Objectives

Aims: Broad statement(s) of intent
Identify the project's purpose

Objectives: Identify specific, measurable achievements
Quantitative and qualitative measures by which completion of the project can be judged

- Expected outcomes/deliverables

Identify what will be produced/submitted in the project

- Keywords

Identify the topic areas that the project draws on

Preparing a project proposal: Explicit Structure (2)

- **Introduction/Background/Overview**

- Overview of the project (Identification of research questions and hypotheses, elaboration of aims)
- Motivation for the project
- Motivation for **you** conducting the project

- **Related Research**

Identifies other work, publications, and related to the same/similar topic

- **Methods**

Identifies the research methods and project methods that will be used (e.g. theoretical investigation, case study)

Preparing a project proposal: Explicit Structure (3)

- Research Requirements

Identifies the resources that will be needed for the project (e.g. hardware, software, data, personnel)

- Project Plan

- More or less detailed 'timetable' for the project
- Deadlines for deliverables

Today's questions

Consider the proposal for an [academic project](#) taken from (Dawson 2005, p. 50).

- 1 What is wrong with it?
- 2 How could it be improved?

(10 minutes group discussion)

Conclusion

- Choosing the right project is an important stage in any project
- There are a number of techniques that can assist you with choosing the right project
- In a project proposal or project specification
 - stick to the required structure and
 - address all the guiding questions as precisely as possible

Further reading:

Sharp et al. (2002) proposes five questions that might help you to choose a project supervisor; see (Dawson 2005; p. 52).

Research Methods in Computer Science

Lecture 20: Project planning (1)

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Department of Computer Science
University of Liverpool

Previously ...

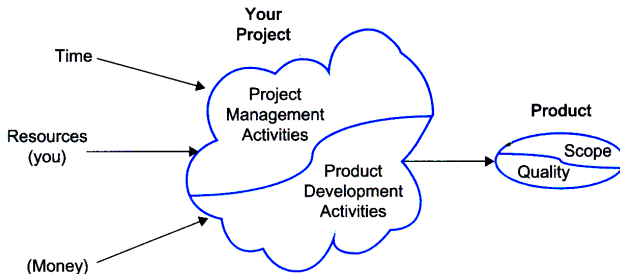
- 38 Project proposals
 - Implicit content
 - Explicit structure
 - Reviewing your proposal

Topics

- 42 Project planning
 - Time estimates
 - Milestones
 - Activity sequencing

Overview

- All **projects** consume **resources** including time and money in order to deliver a **product** of a particular **scope** and **quality**
- There is always a **tension** between the **extent of resource input** and the **extend of product output**
- There is also **tension** between **project management activities** and **project development activities**



Main project activities

Project management

Concerned with

- planning the conduct of the project
- controlling and checking project progress
- monitoring milestones and deliverables
- managing risk

Should account for not more than 10% of overall effort

↪ not evenly distributed; spend most of it towards the start!

'Product' development

Concerned with

- achieving the aims and objectives of the project
- producing the deliverables in accordance with the project plan
- optimising scope and quality of the deliverables relative to the resources available

Project stages

From a **project management** perspective, projects proceed in five stages:

① Definition

Deciding on a project; making a project proposal

② Planning

Detailed planning of the project

③ Initiation

Organising work (in particular, group work); literature survey

④ Control

Monitoring the progress of the project

⑤ Closure

Delivering/deploying result of the project; preparing final presentation; writing up reports

Project definition: Aims and Objectives (1)

Clear specification of what the project is to achieve

→ definition of **aims** and **objectives**

Aims: Broad statement(s) of intent
Identify the project's purpose

Examples:

- Design a methodology for GUI development of technical courseware material
- Develop and evaluate an Artificial Neural Network to predict stock market indices

Project definition: Aims and Objectives (2)

Clear specification of what the project is to achieve

→ definition of **aims** and **objectives**

Example aim:

- Develop and evaluate an Artificial Neural Network to predict stock market indices

Objectives: Identify specific, measurable achievements

Quantitative and qualitative measures by which completion of the project can be judged

Example:

- 1 Complete a literature search and literature review of existing stock market prediction techniques
- 2 Develop a suitable Artificial Neural Network model
- 3 Identify and collect suitable data for analyses and evaluation
- 4 Evaluate the model using appropriate statistical techniques
- 5 Complete final report

Project definition: SMART objectives

Each **objective** should be

- Specific
- Measurable
- Appropriate
- Realistic
- Time-related

Example:

- ① Complete a literature search and literature review of existing stock market prediction techniques
- Is it **specific**? Does it tell us what will be done?
- Is it **measurable**? How will we know to what extent and to what quality the objective has been completed?
- Is it **appropriate**? Does it relate to and in support of our aims?

Project definition: SMART objectives

Each **objective** should be

- Specific
- Measurable
- Appropriate
- Realistic
- Time-related

Example:

- 1 Complete a literature search and literature review of existing stock market prediction techniques
- Is it **realistic**? Can we realistically expect to achieve this objective?
 - Is it **time-related**? Have we identified how long the task will take and when we will complete it?

Project planning

Objectives of project planning

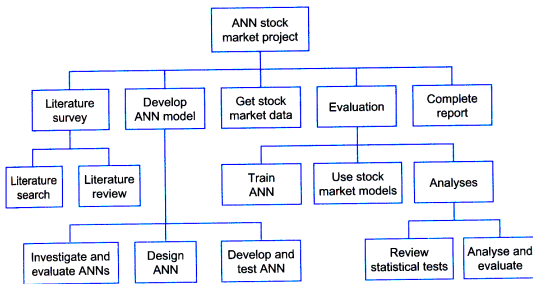
- Identifying the tasks that need to be done
- Clarifying the order in which tasks need to be done
- Determining how long each task will take
- (Redefining the project if there are problems)

Steps of project planning

- 1 Work breakdown
- 2 Time estimates
- 3 Milestone identification
- 4 Activity sequencing
- 5 Scheduling
- 6 Replanning

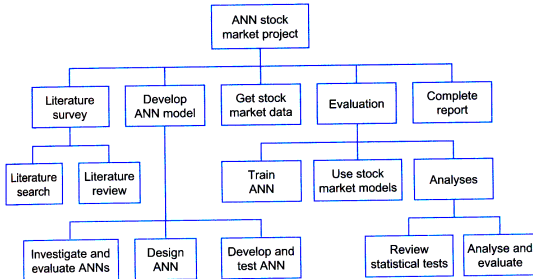
Work breakdown (1)

- First step of project planning: Identifying the tasks that need to be done
- Starting point(s) should be the **objectives** of the project;
Then break your objectives down into lower and lower levels of detail
- **Work breakdown structures** are used to visualise the process of breaking down the project



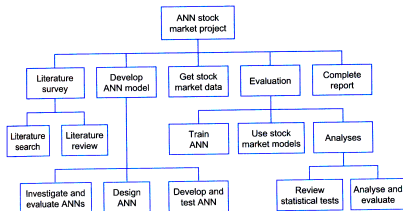
Work breakdown (2)

- Tasks at all levels need to be separate from one another
- Continue to break down your project into smaller tasks until each task takes up no less than 5% of the total effort



Time estimates

- Make reasonably accurate predictions of
 - the **effort** needed for completion and
 - the **duration** until completion
 of each leaf node of the work breakdown structure
- If the estimate exceeds the total time available for the project, then either modify the objectives and work breakdown or reduce and reallocate time between tasks



Activity	Effort	Duration
Literature search	2 weeks	8 weeks
Literature review	2 weeks	4 weeks
Investigate and evaluate ANNs	2 weeks	4 weeks
Design ANN	2 weeks	4 weeks
Develop and test ANN	2 weeks	2 weeks
Get stock market data	1 week	1 week
Train ANN	1 week	1 week
Use stock market models	1 week	2 weeks
Review statistical tests	1 week	2 weeks
Analyse and evaluate	4 weeks	4 weeks
Complete report	8 weeks	8 weeks
Total	26 weeks	40 weeks

Research Methods in Computer Science

Lecture 20: Project planning (2)

Ullrich Hustadt

Department of Computer Science
University of Liverpool

Previously ...

- 36 Introduction
- 37 Project definition
 - Aims and objectives
- 38 Project planning
 - Steps
 - Work breakdown
 - Time estimates

Steps of **project planning**

- 1 Work breakdown
- 2 Time estimates
- 3 Milestone identification
- 4 Activity sequencing
- 5 Scheduling
- 6 Replanning

Topics

- 39 Project planning
 - Time estimates
 - Milestones
 - Activity sequencing

Running example

Example aim:

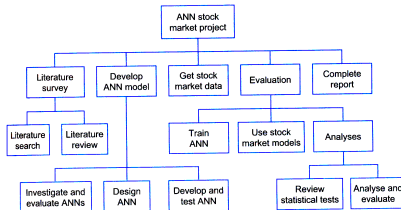
Develop and evaluate an Artificial Neural Network to predict stock market indices

Example objectives:

- 1 Complete a literature search and literature review of existing stock market prediction techniques
- 2 Develop a suitable Artificial Neural Network model
- 3 Identify and collect suitable data for analyses and evaluation
- 4 Evaluate the model using appropriate statistical techniques
- 5 Complete final report

Time estimates

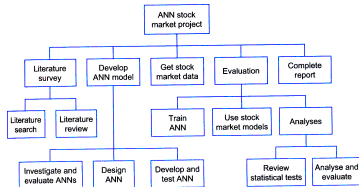
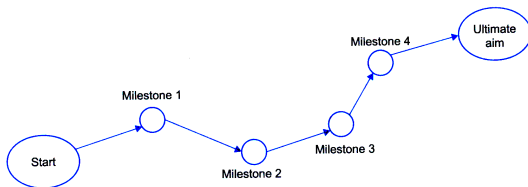
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Analyse and evaluate	4 weeks	4 weeks
Complete report	8 weeks	8 weeks
Total	26 weeks	40 weeks

Milestone identification

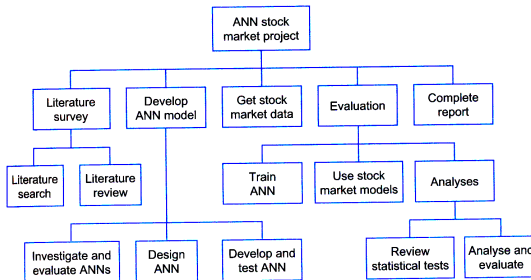
- **Milestones** are significant steps towards the completion of the project
 ~~~ intermediate goals at which to aim



- M1 Completion of literature review  
 (M2 Completion of ANN development)  
 (M3 Completion of evaluation)  
 M4 Completion of project/report

# Activity sequencing

- The work breakdown structure does **not** state in which order tasks are performed

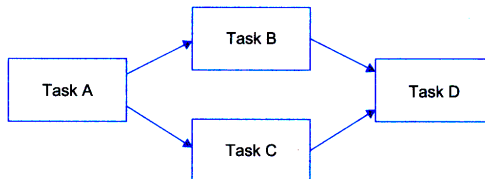


- To represent the order and inter-dependency of tasks we can use **activity networks**
  - Activity-on-the-node diagrams
  - Activity-on-the-arrow diagrams

# Activity-on-the-node diagrams

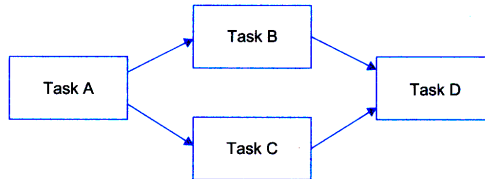
- Tasks are represented by rectangular nodes
- Milestones are represented by diamond-shape nodes
- Arrows indicate the order in which they need to be performed

Example:



- Task A has to be completed before tasks B and C can start
- Task B and C can be done independently (in parallel)
- Task D can only start once both tasks B and C have been completed

# Activity-on-the-node diagrams: Start and end dates



- Assume we estimate effort and duration for the four tasks as follows

| Activity | Effort  | Duration |
|----------|---------|----------|
| Task A   | 2 weeks | 4 weeks  |
| Task B   | 3 weeks | 4 weeks  |
| Task C   | 2 weeks | 4 weeks  |
| Task D   | 2 weeks | 3 weeks  |

- Also assume
  - the project starts on 1 January
  - each month has four weeks
  - there are no breaks, holidays, etc
- What is the start date for each of the tasks?

# Activity-on-the-node diagrams: Critical path

- **Critical path:** Longest-duration path through a network  
~> identifies the tasks in the project that must not be delayed
- Determination of critical paths:
  - Work backwards from the end to the start
  - As long as there is only one preceding task, this task must be on the critical path
  - If there is more than one preceding tasks, only the task(s) which force the start time of the next task are on the critical path
- ~> there can be more than one critical path

# Research Methods in Computer Science

## Lecture 22: Project planning (3)

Ullrich Hustadt

Department of Computer Science  
University of Liverpool

# Previously ...

- 42 Project planning
  - Time estimates
  - Milestones
  - Activity sequencing

## Steps of project planning

- 1 Work breakdown
- 2 Time estimates
- 3 Milestone identification
- 4 Activity sequencing
- 5 Scheduling
- 6 Replanning

# Topics

## 43 Developing an activity diagram

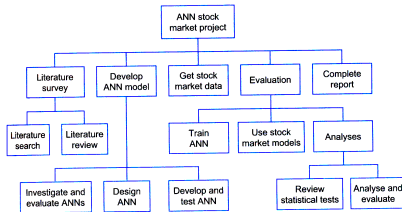
## 44 Project planning

- Problems with ADs
- Scheduling
- Replanning
- Rolling wave planning



# Group work

- Working in groups, construct an activity-on-the-node diagram for the example stock market project based on our example project



| Activity                        | Effort          | Duration        |
|---------------------------------|-----------------|-----------------|
| 1 Literature search             | 2 weeks         | 8 weeks         |
| 2 Literature review             | 2 weeks         | 4 weeks         |
| 3 Investigate and evaluate ANNs | 2 weeks         | 4 weeks         |
| 4 Design ANN                    | 2 weeks         | 4 weeks         |
| 5 Develop and test ANN          | 2 weeks         | 2 weeks         |
| 6 Get stock market data         | 1 week          | 1 week          |
| 7 Train ANN                     | 1 week          | 1 week          |
| 8 Use stock market models       | 1 week          | 2 weeks         |
| 9 Review statistical tests      | 1 week          | 2 weeks         |
| 10 Analyse and evaluate         | 4 weeks         | 4 weeks         |
| 11 Complete report              | 8 weeks         | 8 weeks         |
| <b>Total</b>                    | <b>26 weeks</b> | <b>40 weeks</b> |

## Dependencies

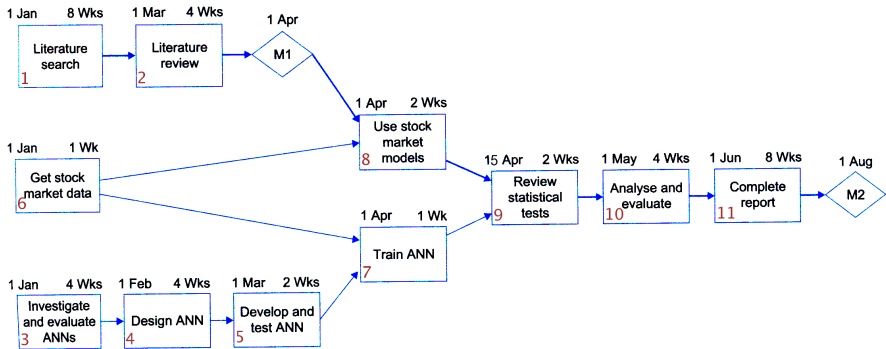
|                 |               |       |
|-----------------|---------------|-------|
| 2 → 1           | 7 → 5 → 4 → 3 | 8 → 6 |
| 11 → 10 → 9 → 8 | 9 → 7 → 6     | 8 → 2 |

## Milestones

|    |                                 |
|----|---------------------------------|
| M1 | Completion of literature review |
| M4 | Completion of project/report    |

- Determine start dates for each task
- Determine the critical path(s) for this project

# Solution



# Problems with activity diagrams

- **Correctness** of activity diagrams is difficult to check

Example:

| Activity | Effort | Duration |
|----------|--------|----------|
| Task A   | 1 week | 4 weeks  |
| Task B   | 1 week | 4 weeks  |

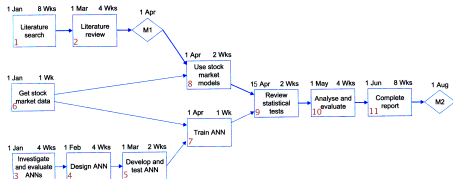
Question: Can tasks A and B done in parallel and both be finished within 4 weeks?

Answer: Information is insufficient to tell

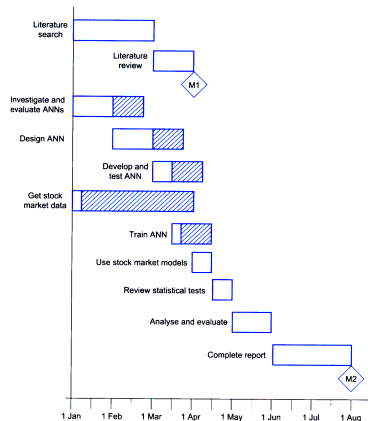
- Do not allow to express **distribution of effort** within a task
- Do not reflect the **duration/effort** of each task well (all nodes are of equal size)
- Do not allow to indicate **slack**
- **Simplistic view of activities/tasks**: No loops, no conditions

# Scheduling

| Activity                      | Effort          | Duration        |
|-------------------------------|-----------------|-----------------|
| Literature search             | 2 weeks         | 8 weeks         |
| Literature review             | 2 weeks         | 4 weeks         |
| Investigate and evaluate ANNs | 2 weeks         | 4 weeks         |
| Design ANN                    | 2 weeks         | 4 weeks         |
| Develop and test ANN          | 2 weeks         | 2 weeks         |
| Get stock market data         | 1 week          | 1 week          |
| Train ANN                     | 1 week          | 1 week          |
| Use stock market models       | 1 week          | 2 weeks         |
| Review statistical tests      | 1 week          | 2 weeks         |
| Analyse and evaluate          | 4 weeks         | 4 weeks         |
| Complete report               | 8 weeks         | 8 weeks         |
| <b>Total</b>                  | <b>26 weeks</b> | <b>40 weeks</b> |

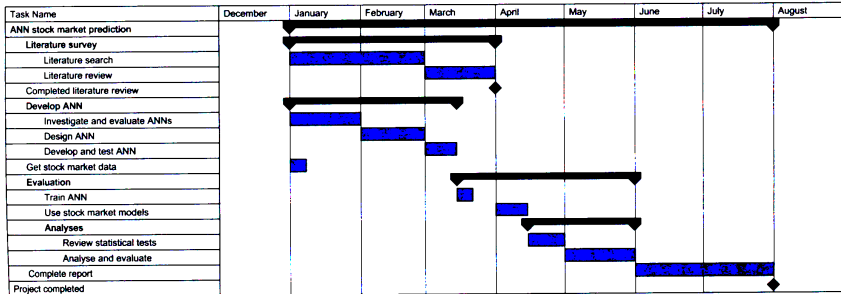


## Gantt Chart



- Activities are represented by rectangles
- Milestones are represented by diamonds
- Size indicates duration relative to the timeline
- Shaded areas indicate **slack**

# MS Project Gantt Chart



- MS Project allows to represent the hierarchy of the **work breakdown structure**
- MS Project allows to represent activities and milestones (in the expected way)
- MS Project does not allow to represent slack
- MS Project does not allow to represent interdependencies across high-level tasks

# Replanning

- Needs to be done if you try to achieve too much in too little time
- **Approach:**
  - Iterate the following steps until you get a correct schedule
    - Rethink the interdependencies between activities
    - Redo estimates for effort and duration of each tasks
    - Reschedule tasks
    - Rethink the aims and objectives of your project
    - Redo work breakdown structure
- No plan is perfect; no plan is set in stone

# Rolling wave planning

- Phased iterative approach to project planning  
    ~> fits well for incremental development
- Approach:
  - 1 Define **planning packages** for your project with
    - resource requirements
    - macro level deliverables
    - macro level dependencies
  - 2 Execute the following loop
    - 1 Determine which planning package has to be done next (first)
    - 2 Make a detailed plan for this planning package
    - 3 Execute the plan
    - 4 Re-adjust the remaining planning packages based on what happened

# Research Methods in Computer Science

## Lecture 23: Risk management

Ullrich Hustadt

Department of Computer Science  
University of Liverpool



# Previously ...

## 43 Developing an activity diagram

## 44 Project planning

- Problems with ADs
- Scheduling
- Replanning
- Rolling wave planning

### Steps of project planning

- 1 Work breakdown
- 2 Time estimates
- 3 Milestone identification
- 4 Activity sequencing
- 5 Scheduling
- 6 Replanning

# Topics

## 45 Risk management

- Introduction
- Identify risks
- Assess impact of risks
- Alleviate critical risks
- Control risks

## 46 Project planning: Summary

# Risk management: Introduction

## Risk management

- involves the identification of risks at the project's outset
- control of those risks as the project progresses

→ risk management process

Four main stages of the risk management process

- 1 Identify risks
- 2 Assess impact of risks
- 3 Alleviate critical risks
- 4 Control risks

# Identifying risks: Types of risk

|               | Event-driven                                       | Evolving                                                                                         |
|---------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Technical     | Project requirements change;<br>Hard disk crashing | Project beyond your technical capability;<br>Problem dependent on developing a complex algorithm |
| Non-Technical | Supervisor leaving;<br>Illness                     | Underestimating effort required for a task;<br>Literature not arriving on time                   |

# Identifying risks

## Risk triggers (risk symptoms)

Events happening during the course of a project that might indicate problems or that one of the identified risks is increasingly likely to occur

### Examples:

- Missing preliminary milestones in your project
- Struggling with a straightforward implementation of a component
- Problems with arranging a meeting a client

# Assessing the impact of risks (1)

Risk impact = Likelihood  $\times$  Consequence

Example: Severe earthquakes in Britain

- Likelihood is low
- Residential building  $\rightarrow$  Consequences are low  
Nuclear power plant  $\rightarrow$  Consequences are catastrophic
- ~ Nuclear power plants are earthquake proof,  
residential buildings are not

# Assessing the impact of risks (2)

- 1 Assess each risk according to the following scales:

| Risk Likelihood | Score |
|-----------------|-------|
| Low             | 1     |
| Medium          | 2     |
| High            | 3     |

| Risk Consequence | Score |
|------------------|-------|
| Very low         | 1     |
| Low              | 2     |
| Medium           | 3     |
| High             | 4     |
| Very high        | 5     |

- 2 Compute risk impact for each risk using the formula

$$\text{Risk impact} = \text{Likelihood} \times \text{Consequence}$$

- 3 Rank all risks according to their risk impact

# Assessing the impact of risks (3)

## 4 Determine critical risks

### (a) 80/20 rule:

20% of your risks cause 80% of your problems

~ 20% top ranking risks are critical

### (b) RAG grading:

Red Risks with impact greater than 10

~ critical risks

Amber Risks with impact between 6 and 10

~ deserve some attention

Green Risks with impact smaller than 6

~ can be ignored



# Alleviating critical risks (1)

- Contingency

Accepting that the risk is going to occur and putting something in place to deal with it when it does

Examples:

- Hard disk crash → have a backup
- Time over-run → allow slack for each task

- Deflection

Passing the risk on to someone or something else

Example:

- Required software → use of existing software instead of developing it yourself

# Alleviating critical risks (2)

- Avoidance

Reducing the likelihood that the risk will occur at all

## Examples:

- Use of programming languages
  - use one that you know instead of one that you don't
- Development of a complex algorithm
  - modify an existing algorithm

# Risk assessment report

Project: Introduction of IT-assisted teaching at a college

| Risk                                               | Likelihood | Consequence  | Risk management approach                                                  | Risk symptoms                               |
|----------------------------------------------------|------------|--------------|---------------------------------------------------------------------------|---------------------------------------------|
| Infrastructure                                     |            |              |                                                                           |                                             |
| IT infrastructure cannot cope with requirements    | Med(2)     | High(4)      | Equip sufficiently and involve IT Dept                                    | Speed of equipment response                 |
| Data projector failing during teaching             | Low(1)     | Very High(5) | Have a stand-by data projector                                            | None                                        |
| Staff                                              |            |              |                                                                           |                                             |
| Lack of commitment by staff                        | Med(2)     | High(4)      | Clear communication plan; staff development events                        | Non- or variable attendance of events       |
| Loss of key staff                                  | Med(2)     | Med(3)       | Succession planning; critical procedures should be documented in a manual | Notice period / Request to attend interview |
| Delivery                                           |            |              |                                                                           |                                             |
| Staff not available at times training is delivered | High(3)    | High(4)      | Flexible delivery and session on different days and at different times    | Timetables                                  |

# Controlling risks

## Planning a risk strategy

- How will you go about managing/controlling the risks identified?

E.g. how and when would you notice a time over-run?

**Checkpoints:** Checking critical risks

- at regular intervals (e.g. weekly)
  - at the end of particular project stages
  - at meetings with your supervisor
- 
- How and when will you check the risk triggers identified?
  - How and when will you invoke your contingency plans?
  - How and when will you update your critical risk list?

Risk likelihood and risk consequences change over time

# Today's question

Consider our running example, that is, the project with the aim to

*Develop and evaluate an Artificial Neural Network to predict stock market indices*

which is conducted by undertaking the following tasks

| Activity                      | Effort          | Duration        |
|-------------------------------|-----------------|-----------------|
| Literature search             | 2 weeks         | 8 weeks         |
| Literature review             | 2 weeks         | 4 weeks         |
| Investigate and evaluate ANNs | 2 weeks         | 4 weeks         |
| Design ANN                    | 2 weeks         | 4 weeks         |
| Develop and test ANN          | 2 weeks         | 2 weeks         |
| Get stock market data         | 1 week          | 1 week          |
| Train ANN                     | 1 week          | 1 week          |
| Use stock market models       | 1 week          | 2 weeks         |
| Review statistical tests      | 1 week          | 2 weeks         |
| Analyse and evaluate          | 4 weeks         | 4 weeks         |
| Complete report               | 8 weeks         | 8 weeks         |
| <b>Total</b>                  | <b>26 weeks</b> | <b>40 weeks</b> |

What might a **risk assessment report** look like for this project?

(10 minutes group discussion)

# Project planning: Summary

- **Project planning** consists of two stages:
  - 1 Defining what it is you want to achieve
  - 2 Planning how you will achieve it
- **Project planning** proceeds in six steps
  - 1 Work breakdown
  - 2 Time estimates
  - 3 Milestone identification
  - 4 Activity sequencing
  - 5 Scheduling
  - 6 Replanning
- **Risk management** is performed in parallel with project management and involves four stages:
  - 1 Risk identification
  - 2 Risk quantification
  - 3 Risk alleviation
  - 4 Risk control

# Research Methods in Computer Science

## Lecture 24: Legal, Social, Ethical and Professional Issues

Ullrich Hustadt

Department of Computer Science  
University of Liverpool

# Previously ...

## 45 Risk management

- Introduction
- Identify risks
- Assess impact of risks
- Alleviate critical risks
- Control risks

## 46 Project planning: Summary



# Topics

Module aim:

To remind students of the Legal, Social, Ethical and Professional (LSEP) issues applicable to the computer industry.

## 47 (Social) Context

## 48 Ethics

- Overview
- Theories
- Relativism
- Divine
- Kantianism
- Utility
- Social contract

# Context

- Context of your **work** and **life** involves
  - Cultural
  - Social
  - Legal
  - Ethicalaspects
- These in turn depend on your
  - individual
  - societalsituation/view

# Social context

- Ownership
- Participation
- Employment
- Entertainment
- Government
- Security
- Privacy

# Morality and Ethics

## Quinn (2004)

Every society has rules of conduct describing what people ought and ought not to do in various situations. We call these rules morality. Ethics is the philosophical study of morality, a rational examination into peoples moral beliefs and behaviour.

Michael J. Quinn: Ethics for the Information Age.  
Addison Wesley, 2004.

# Key ethical theories

- 1 Subjective relativism
- 2 Cultural relativism
- 3 Divine command
- 4 Kantianism
- 5 Act utilitarianism
- 6 Rule utilitarianism
- 7 Social contract theory

# Relativism

- Maintains that there are no universal moral norms of right or wrong
- **Subjective relativism** holds that each person decides for themselves
- Problems:
  - Impossible to argue about morality
  - No universal morality
- **Cultural relativism** holds that society decides what is right or wrong
- Problems:
  - How exactly does a society decide morality
  - No universal morality

# Divine command

- Morality emerges as behaviour in line with the will of God
- In the context of **cultural relativism** explains where morality comes from
- Problems:
  - How exactly do we get to know God's will
  - Makes it difficult to argue about morality
  - No universal morality

# Kantianism

- Based on the work of Immanuel Kant (1724–1804)
- Personal morality ought to be guided by principles which are universal
- He formulated the following **categorical imperative**:

*Act only according to that maxim by which you can at the same time will that it would become a universal law.*

- Called **categorical imperative** because
  - it is an absolute, unconditional requirement that exerts its authority in all circumstances, and
  - it is both required and justified as an end in itself.



# Kantianism (2)

Kant divides the duties imposed by this formulation into two subsets:

## Perfect duty:

Do not act by maxims that result in logical contradictions when we attempt to universalise them.

**Example:** Lying

→ destroys the meaningfulness of language

## Imperfect duty:

Act only by maxims that we would desire to be universalised.

**Example:** Only help another person if there is something in for yourself

→ no one ever wants selfless help from another person

# Kantianism (3)

- Pros:

- Rational
- Produces universal moral guidelines
- All people are treated as moral equals

- Cons:

- Has problems with conflicting rules

Example: You should not steal!

You should feed your children!

You should (not) steal to feed your children?

- No way to resolve a conflict between rules
- No exceptions to moral laws
- Leaves very little room for personal freedom

# Utilitarianism (1)

- Due to Jeremy Bentham (1748–1832) and John Stuart Mill (1806–1873)
- **Goodness** is ‘benefit’ as opposed to ‘harm’
- The greatest good (or happiness) is the principle of utility and should be the aim of personal and social morality
- We can distinguish

## Act utilitarianism

An action is good if its net effect, over all those affected, is likely to produce more happiness than unhappiness

## Rule utilitarianism

Those moral rules should be adopted that if followed by everyone will lead to the greatest increase in total happiness

# Utilitarianism (2)

- Pros:

- Can be applied to acts as well as rules
- It is comprehensive
- It is easy to understand

- Cons:

- How do you define 'happiness'?
- What if we cannot agree on a definition?
- Who is included in the computation of total happiness?
- Is an unequal distribution of happiness fair?

# Social contract (1)

- Proposed by Thomas Hobbes (1603–1679) and Jean-Jacques Rousseau (1712–1778)
- Moral rules are motivated by the conflict between our desire of **freedom** and our desire of **security**
- By joining together through a **social contract** and abandoning their claims of natural right, individuals can both preserve themselves and remain free

*Morality consists in the set of rules, governing how people treat one another, that rational people will agree to accept, for their mutual benefit, on the condition that others follow those rules as well (Rachels, 2003; as quoted by Quinn, 2004).*

# Social contract (2)

- Pros:

- Framed in the language of (individual) rights
- Explains selfish actions in the absence of common agreement
- Explains the relationship between people and government

- Cons:

- How exactly do you set up a social contract?
- How do you become subject to a social contract?
- How do you enforce a social contract?
- How do we resolve conflicting rights?

# Comparison table

| Theory              | Motivation       | Criteria      | Focus      |
|---------------------|------------------|---------------|------------|
| Kantianism          | Duty             | Actions/Rules | Individual |
| Act utilitarianism  | Consequence      | Actions       | Group      |
| Rule utilitarianism | Consequence/Duty | Rules         | Group      |
| Social contract     | Rights           | Rules         | Individual |

# Ethical theories: Summary

- Used to explain how and why we should act
- Determine what is or is not ethical
- Relates to individual and the group
- Key ethical theories
  - 1 Subjective relativism
  - 2 Cultural relativism
  - 3 Divine command
  - 4 Kantianism
  - 5 Act utilitarianism
  - 6 Rule utilitarianism
  - 7 Social contract theory



# Research Methods in Computer Science

## Lecture 25: Legal, Social, Ethical and Professional Issues

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# Previously ...

Module aim:

To remind students of the Legal, Social, Ethical and Professional (LSEP) issues applicable to the computer industry.

## 47 (Social) Context

## 48 Ethics

- Overview
- Theories
- Relativism
- Divine
- Kantianism
- Utility
- Social contract

# Topics

- 49 Professional Ethics
- 50 BCS Code of Conduct
- 51 ACM Code of Conduct

# Professional Ethics

- More restrictive than personal ethics because it involves the more restrictive society/culture of work and commerce
- Applies also to other restrictive social contracts such as study
- Many flavours of professional ethics exist: medical, engineering, banking, etc
- Often associated with formal structure  
BMC (medicine), IEE (engineering), BCS (computing)
- Often formally constructed rules and codes of conduct  
Hippocratic oath taken by doctors

# The IT Professional Context

- Legal obligations
- Social obligations
- Ownership/sharing obligations
- IP obligations
- Product development process obligations
- Product quality obligation
  - validity
  - robustness
  - simplicity
  - modifiability
  - reusability
  - compatibility
  - efficiency
  - ease of use
  - portability
  - integrity
- Product consequence obligations

# BCS Code of Conduct

Four component parts:

- 1 The public interest
- 2 Duty to employers and clients
- 3 Duty to the profession
- 4 Professional integrity and competence

Associated with these four components is an injunction to be aware that changes happen and change should be monitored

# The Public Interest

- Safeguarding public health and safety
- Have regards to the rights of third parties
- Understand and comply with relevant legislation
- Understand and respect issues of human rights

# Duty to employers and clients

- Act professionally
- Respect the privacy and rights of employers and clients
- Be honest about products and services



# Duty to profession

- Act so as to promote
  - general awareness of IT issues
  - conformance to code of conduct
- Act with integrity towards other professionals
- Do not make public statements if you are unqualified to do so

# Professional competence and integrity

- Upgrade and maintain knowledge of the field
- Understand and conform to current best practices
- Avoid conflicts of interest

# ACM Code of Conduct

- General moral imperatives
- Professional responsibilities
- Organisational leadership
- Compliance with the code

# General moral imperatives

- Contribute to society and human well-being
- Avoid harm to others
- Be honest and trustworthy
- Be fair and take action not to discriminate
- Honor property rights including copyrights and patents
- Give proper credit for intellectual property
- Respect the privacy of others
- Honor confidentiality

# Professional responsibilities

- Strive to achieve the highest quality, effectiveness and dignity in both the process and products of professional work
- Acquire and maintain professional competence
- Know and respect existing laws pertaining to professional work
- Accept and provide appropriate professional review
- Give comprehensive and thorough evaluation of computer systems and their impacts, including analysis of possible risks
- Honor contracts, agreements, and assigned responsibilities
- Improve public understanding of computing and its consequences
- Access computing and communication resources only when authorised to do so

# Organisational leadership

- Articulate social responsibilities of members of an organisational unit and encourage full acceptance of those responsibilities
- Manage personnel and resources to design and build information systems that enhance the quality of working life
- Acknowledge and support proper and authorised uses of an organisation's computing and communication resources
- Ensure that users and those who will be affected by a system have their needs clearly articulated during the assessment and design of requirements; later the system must be validated to meet those requirements
- Articulate and support policies that protect the dignity of users and others affected by a computing system
- Create opportunities for members of the organisation to learn the principles and limitations of computer systems

# Compliance with the code

- Uphold and promote the principles of this code
- Treat violations of this code as inconsistent with membership in the ACM

# Research Methods in Computer Science

## Lecture 26: Legal, Social, Ethical and Professional Issues

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University of Liverpool



# Previously ...

Module aim:

To remind students of the Legal, Social, Ethical and Professional (LSEP) issues applicable to the computer industry.

- 49 Professional Ethics
- 50 BCS Code of Conduct
- 51 ACM Code of Conduct

# Topics

## 52 Standards and Standardisation

- Standards Organisations
- ISO
- IETF
- ITU

## 53 Legal issues

- Copyright
- Computer Misuse
- Software Audit
- Data Protection
- Disability Discrimination
- Freedom of Information

# Standards and Standardisation

- Are written agreements on some technical matter that seeks to ensure that what is governed is fit for purpose
- Standards may be
  - Personal
  - Professional
  - Organisational
  - Social
- Care should be taken to note who is standardising what, and for what purpose

# Standards Organisations

- **ISO** — International Standards Organisation <http://www.iso.org/>
- **ANSI** — American National Standards Institute  
<http://www.ansi.org/>
- **CEN** — Comité Européen de Normalisation  
<http://www.cenorm.be/cenorm/>
- **BSI** — British Standards Institute  
<http://www.bsonline.bsi-global.com>
- **DIN** — Deutsches Institut für Normung  
<http://www.din.de/>
- **IETF** — Internet Engineering Task Force <http://www.ietf.org/>
- **ITU** — International Telecommunication Union  
<http://www.itu.int/home/>

# ISO: International Standards Organisation

- A federation of national standards bodies from some 130 countries
- Non-governmental
- Established in 1947
- ISO's mission

To promote the development of standardisation and related activities in the world with a view to facilitating the international exchange of goods and services, and to developing cooperation in the spheres of intellectual, scientific, technological and economic activity

- ISO's work results in international agreements which are published as International Standards
- Among those are standards for **programming languages** (C, C++, Ada) and **processes** (quality assurance ISO 9001)

# IETF: Internet Engineering Task Force

- International community of network designers, operators, vendors, and researchers; open to any interested individual
- First IETF meeting was held in 1986
- IETF's mission

The mission of the IETF is to produce high quality, relevant technical and engineering documents that influence the way people design, use, and manage the Internet in such a way as to make the Internet work better. These documents include protocol standards, best current practices, and informational documents of various kinds.

- Every IETF standard is published as an RFC 'Request For Comments'

# ITU: International Telecommunication Union

- International organisation of
  - member states (189)
  - sector members (602)  
e.g. phone companies, TV companies, equipment manufactures
- Traces its origins back to 1865
- ITU's mission is to coordinate the operation of telecommunication networks and services and advance the development of communications technology
- Works through conferences and meetings, where members negotiate the agreements which serve as the basis for the operation of global telecommunication services, based on specifications and recommendations prepared by experts

IETF and ITU are 'in conflict' concerning who is responsible for the future of the Internet and related communications technology

# Legal issues

- Copyright, Designs and Patents Act, 1988
- Computer Misuse Act, 1990
- Data Protection Act, 1984, 1998
- Disability Discrimination Act, 1995
- Freedom of Information Act, 2001
- The Copyright and Related Rights Regulations 2003



# Copyright, Designs and Patents Act, 1988

Copyright is a **property right** which applies to

- original literary, dramatic, musical or artistic works

**literary work** covers computer programs, song lyrics

**dramatic work** covers work of dance or mime

**musical work** covers music exclusive of lyrics, dance, mime

- sound recordings, films, broadcasts or cable programmes

**sound recording** is a recording of sounds, from which the sounds may be reproduced, regardless of the medium

- the typographical arrangement of published editions

Copyright lies with the **author** of a work, or, if the work has been created by an employee in the course of his employment, with the **employer**

# Copyright, Designs and Patents Act, 1988

## States

*The owner of the copyright has exclusive right to copy the work*

It follows that it is illegal to

- copy or distribute software or its documentation without the permission or licence of the copyright owner
- run purchased software on two or more computer simultaneously unless the licence specifically allows it
- knowingly or unknowingly to allow, encourage or pressure employees to make or use illegal copies within an organisation
- infringe laws against unauthorised software copying because a superior, colleague or friend compels or requests it
- loan software in order that a copy be made of it

# Copyright, Designs and Patents Act, 1988

Other stipulations of the act deal with

- importing infringing copy

The **copyright** in a work is infringed by a person who, without the licence of the copyright owner, **imports into the United Kingdom, otherwise than for his private and domestic use**, an article which is, and which he knows or has reason to believe is, an infringing copy of the work

- providing means for making infringing copies

Copyright in a work is infringed by a person who, without the licence of the copyright owner (a) **makes**, (b) **imports into the United Kingdom**, (c) **possesses in the course of a business**, or (d) **sells or lets for hire, or offers or exposes for sale or hire**, an article specifically designed or adapted for making copies of that work, knowing or having reason to believe that it is to be used to make infringing copies

# Copyright, Designs and Patents Act, 1988

Other stipulation of the act deals with copyright work which has an electronic form of is copy-protection.

- devices designed to circumvent copy-protection

The person issuing the copies to the public has the same rights against a person who, knowing or having reason to believe that it will be used to make infringing copies (a) makes, imports, sells or lets for hire, offers or exposes for sale or hire, or advertises for sale or hire, any device or means specifically designed or adapted to circumvent the form of copy-protection employed, or (b) publishes information intended to enable or assist persons to circumvent that form of copy-protection, as a copyright owner has in respect of an infringement of copyright.

# Copyright, Designs and Patents Act, 1988

- Patents in the UK cannot describe **algorithms** or **mathematical methods**, these are **discoveries**
- Over 15k software patents in the USA, several thousand more issued each year
- Intended to prevent others from using some programming technique
- Several infamous patents for software techniques that most experienced programmers consider trivial (e.g. using XOR to plot on a bitmap display)

# Computer Misuse Act, 1990

Defines three offences:

(1) **Unauthorised access to computer material**

Occurs if

- (a) someone causes a computer to perform any function with intent to secure access to any program or data held in any computer,
- (b) the access he intends to secure is unauthorised, and
- (c) he knows at the time when he causes the computer to perform the function that that is the case

(2) **Unauthorised access with intent to commit or facilitate commission of further offences**

Like (1) but with the intent to commit an offence or to facilitate the commission of such an offence

# Software Audit: Accountability

- In mid-size and large business and organisation [copyright law](#) and [license terms](#) can inadvertently be violated
- To ensure that this does not occur, [software audits](#) are a common approach

## Software audits

- are a regular investigation of the software installed on all computers in an organisation
- ensure that it is authorised or licensed
- minimise the risk of prosecution for software theft
- minimise the risk of viruses through uncontrolled software copying
- ensure technical support is available to all users

# Computer Misuse Act, 1990

Defines three offences:

## (3) Unauthorised modification of computer material

Occurs if

- (a) someone performs any act which causes an unauthorised modification of the contents of any computer; and
- (b) at the time when he does the act he has the requisite intent and the requisite knowledge



# Data Protection Act, 1998

- Repeals an earlier 1984 act

- Defines

**data** is information which can be electronically processed or is recorded as part of a relevant filing system

**personal data** means data which relate to a living individual who can be identified from that data (possibly together with other information)

**sensitive personal data** is personal data relating to race, ethnicity, political opinions, etc.

- Regulates who may keep it, how, and who as access, especially the data subject
- Defines registration of **data controllers** and the role of **audit**
- Defines exemptions, charges and penalties, disclosure

# Disability Discrimination Act, 1995

**Disability:** a physical or mental impairment which has a substantial and long-term adverse effect on his ability to carry out normal day-to-day activities

- The Act requires a provider of services to make **reasonable adjustments** to services, practices, policies, and procedures if it would otherwise impossible or difficult for a disabled person to make use of a service
- The Act makes it unlawful for a provider of services to discriminate against a disabled person
  - (a) in refusing to provide, or deliberately not providing, a service provided to other members of the public;
  - (b) in the standard of service which he provides to the disabled person or the manner in which he provides it to him; or
  - (c) in the terms on which he provides a service to the disabled person.

# Freedom of Information Act, 2001

- Obliges **public authorities and publicly-owned companies** to provide information:
  - through a Publication Scheme
  - in response to requests made under the general right of access
- A **publication scheme** is a public commitment to make certain information available and a guide to how that information can be obtained
- Any person making a request for information to a public authority or publicly-owned company is entitled
  - (a) to be informed in writing by the public authority whether it holds information of the description specified in the request
  - (b) if that is the case, to have that information communicated to him
- There are many exemptions